第10章 常用时序逻辑电路及其应用

内容概要

10.1 寄存器

10.2 计数器

10.1 寄存器

- 寄存器是计算机的一个重要部件，用于暂时存放一组二值代码，如参加运算的数据、运算结果、指令等。它被广泛用于各类数字系统和数字计算机中。
- 由有记忆功能的触发器组成；还有一些接收数据的控制门，以便在同一个接收命令作用下使各触发器同时接收数据。
- 一个触发器能储存1位二值代码，用N个触发器组成的寄存器能储存一组N位的二值代码
- 触发器的触发方式决定了寄存器的触发方式。数码寄存器常用的是上升沿触发的D型触发器（边沿触发）和电位触发器，较少采用主-从触发器。
- 寄存器的操作：读/写/复位（清零）

寄存器
- 数码寄存器
  - 数据寄存器（多位边沿触发器组成）
  - 数据锁存器（多位电位触发器组成）
  - 寄存器阵列
- 移位寄存器
  - 单向移位寄存器
  - 双向移位寄存器

10.1.1 数码寄存器

各种类型的触发器都具有置0、置1和保持（记忆）功能，它们都可以用来构成寄存器，而用**D触发器**和**D锁存器**构成寄存器最为方便。

❖ 工作原理

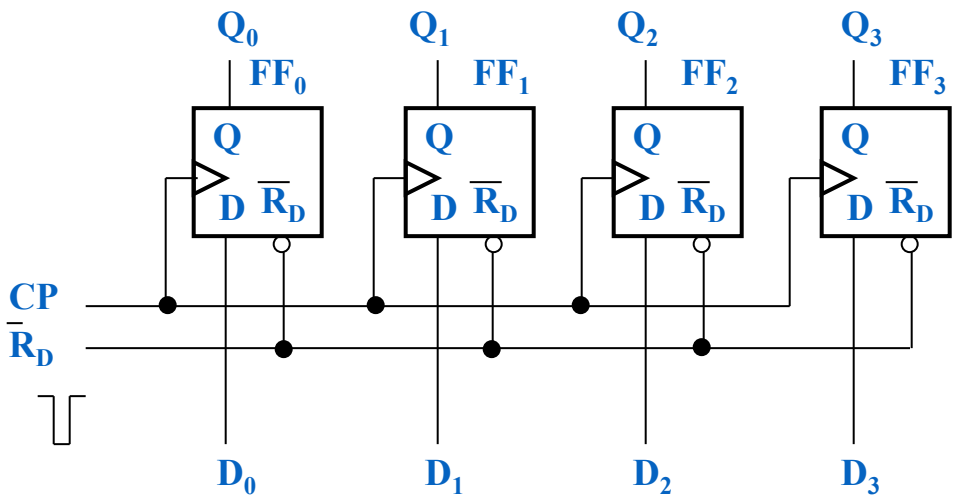
（1）清除（复位）

当 $\overline{R_D}=0$， $Q_0Q_1Q_2Q_3=0000$;

（2）置数（复位端无效时）

当 $CP=CP\uparrow$， $Q_0Q_1Q_2Q_3=D_0D_1D_2D_3$;

❖ 工作方式（数据输入输出方式）——只能并入并出



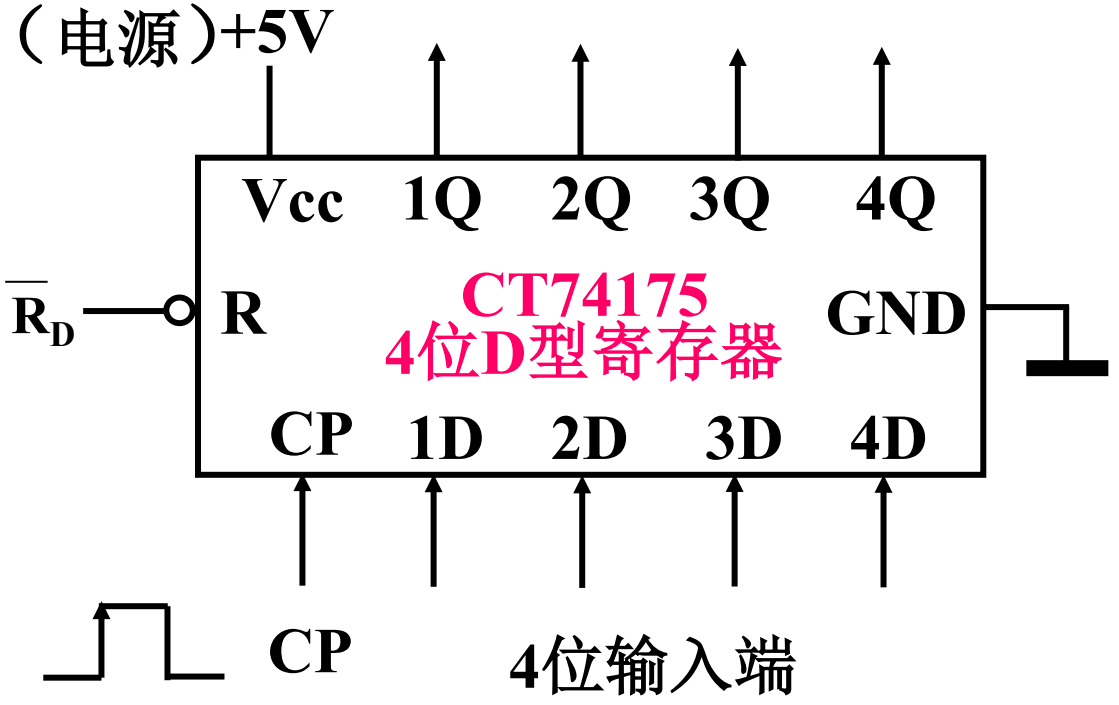
4位D型寄存器

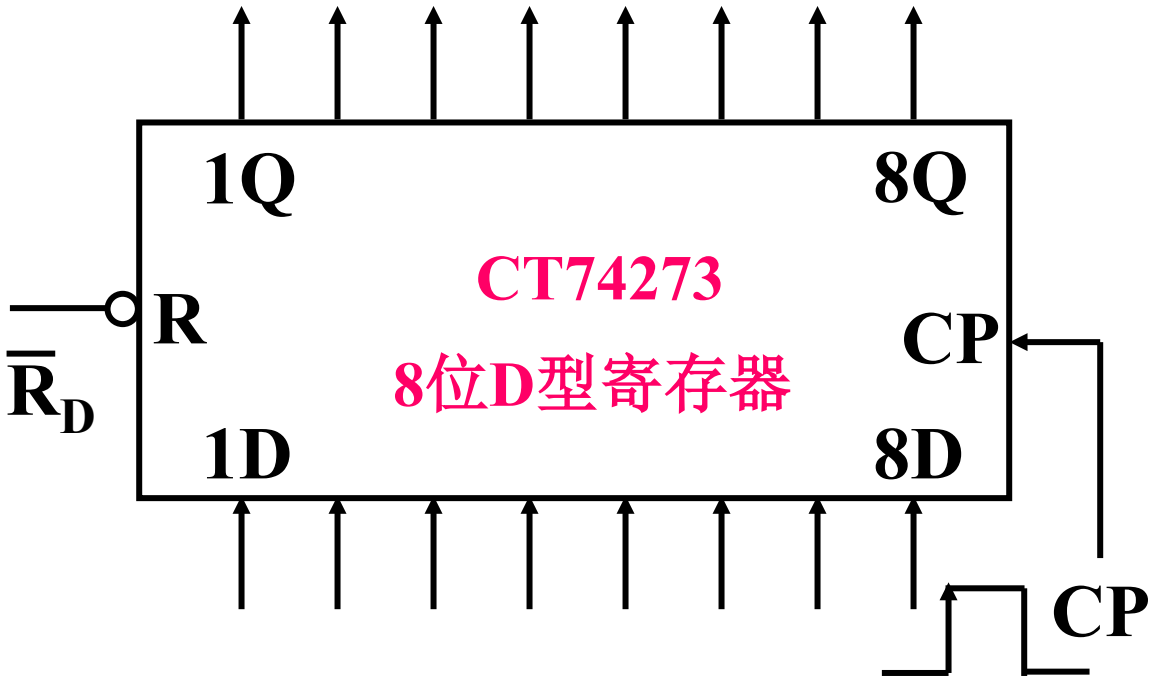
常见的集成数码寄存器

1、TTL型4位D型寄存器：CT74173、CT74175

2、TTL型8位D型寄存器：CT74273、CT74373

3、CMOS型4位D型寄存器：CC4076





M74LS273P

OCTAL POSITIVE EDGE-TRIGGERED D-TYPE FLIP FLOP WITH RESET

DESCRIPTION

The M74LS273P is a semiconductor integrated circuit containing 8 D-type positive edge-triggered flip-flop circuits with common direct reset and clock inputs.

FEATURES

- Positive edge-triggering
- High mounting density with 8 circuits contained
- Direct reset and clock inputs common to all 8 circuits
- Wide operating temperature range ($T_a = -20 \sim +75°C$)

APPLICATION

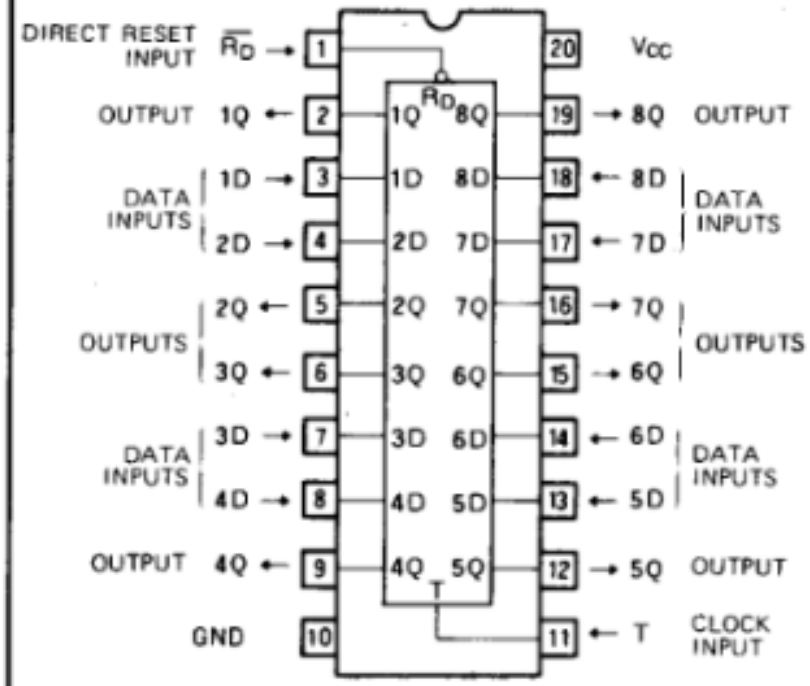
General purpose, for use in industrial and consumer equipment.

FUNCTIONAL DESCRIPTION

This device contains 8 edge-triggered D-type flip-flop circuits and it is provided with direct reset $\overline{R_D}$ input and clock input T common to all 8 circuits. When T changes in each flip-flop from low to high, the data input signal D immediately before the change appears in output Q.

When $\overline{R_D}$ is set low, 1Q through 8Q are all set low irrespective of the status of the ID through 8D and T signals. For use as a D-type flip-flop, $\overline{R_D}$ must be kept in high.

PIN CONFIGURATION (TOP VIEW)



Outline 20P4

https://pdf1.alldatasheet.com/datasheet-pdf/view/1618/MITSUBISHI/74273.html

SN54174, SN54175, SN54LS174, SN54LS175, SN54S174, SN54S175, SN74174, SN74175, SN74LS174, SN74LS175, SN74S174, SN74S175
HEX/QUADRUPLE D-TYPE FLIP-FLOPS WITH CLEAR

- '174, 'LS174, 'S174 Contain Six Flip-Flops with Single-Rail Outputs
- '175, 'LS175, 'S175 Contain Four Flip-Flops with Double-Rail Outputs

description

These monolithic, positive-edge-triggered flip-flops utilize TTL circuitry to implement D-type flip-flop logic. All have a direct clear input, and the '175, 'LS175, and 'S175 feature complementary outputs from each flip-flop.

Information at the D inputs meeting the setup time requirements is transferred to the Q outputs on the positive-going edge of the clock pulse. Clock triggering occurs at a particular voltage level and is not directly related to the transition time of the positive-going pulse. When the clock input is at either the high or low level, the D input signal has no effect at the output.

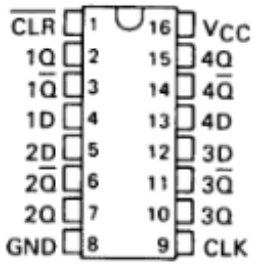
These circuits are fully compatible for use with most TTL circuits.

FUNCTION TABLE
(EACH FLIP-FLOP)

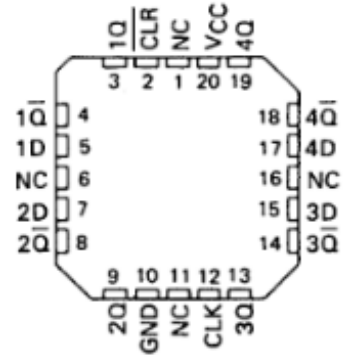
INPUTS			OUTPUTS	
CLEAR	CLOCK	D	Q	$\overline{Q}$†
L	X	X	L	H
H	↑	H	H	L
H	↑	L	L	H
H	L	X	Q_0	$\overline{Q}_0$

H = high level (steady state)
L = low level (steady state)
X = irrelevant
↑ = transition from low to high level
Q_0 = the level of Q before the indicated steady-state input conditions were established.
† = '175, 'LS175, and 'S175 only

SN54175, SN54LS175, SN54S175 . . . J OR W PACKAGE
SN74175 . . . N PACKAGE
SN74LS175, SN74S175 . . . D OR N PACKAGE
(TOP VIEW)

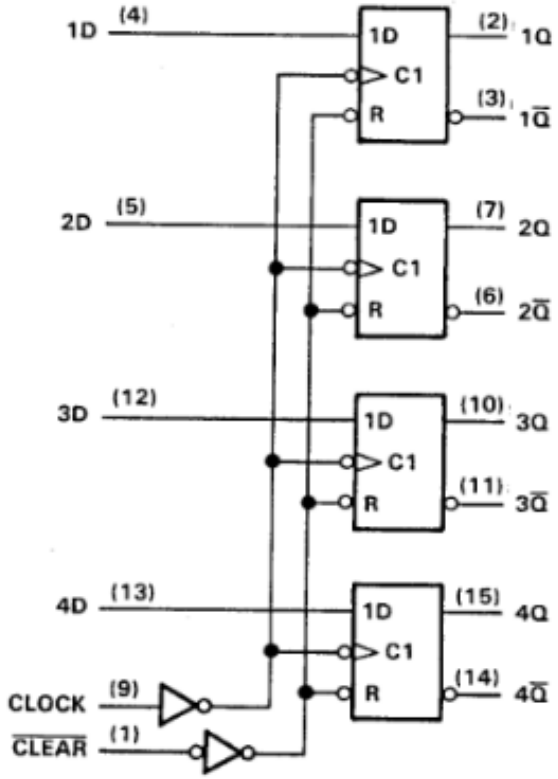


SN54LS175, SN54S175 . . . FK PACKAGE
(TOP VIEW)



NC – No internal connection

'175, 'LS175, 'S175

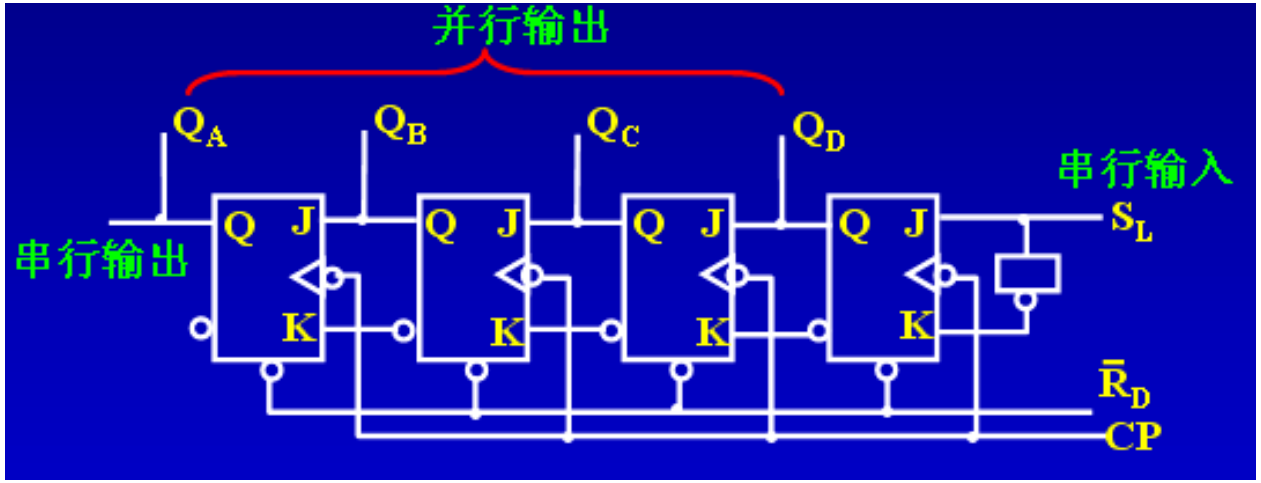


https://pdf1.alldatasheet.com/datasheet-pdf/view/96364/TI/74175.html

10.1.2 移位寄存器

- 具有移位功能的寄存器称为移位寄存器，每来一个时钟脉冲，寄存器中数据就依次向左或向右移一位。
- 在计算机中，常要求寄存器有“移位”功能。例如，移位相加乘法器进行乘法运算时，要求将乘数右移，被乘数左移；除法运算时，要求将余数左移；将并行传递的数转换成串行数据以及将串行传递的数转换成并行数据的过程中，需要移位。
- 分类
  - 左移移位寄存器，右移移位寄存器，双向移位寄存器
- 数据输入方式
  - 串行输入，并行输入
- 数据输出方式
  - 串行输出：右移寄存器、左移寄存器
  - 并行输出：全部触发器的输出作为电路的输出
- 根据数据输入/输出方式，移位寄存器的工作方式有
  - 串入串出、串入并出、并入串出、并入并出

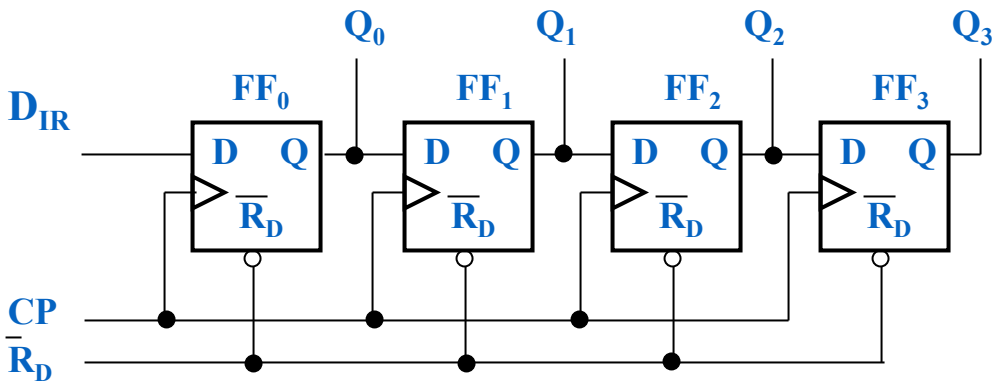
串行输入、串/并行输出单向移位寄存器（1）



左移移位寄存器的串行输入方式：在同一个时钟的控制下，数据从寄存器的最右端串行输入，同时已存入的信息左移一位。

串行输入、串/并行输出单向移位寄存器（2）

D_{IR}：右移串行数据输入1011



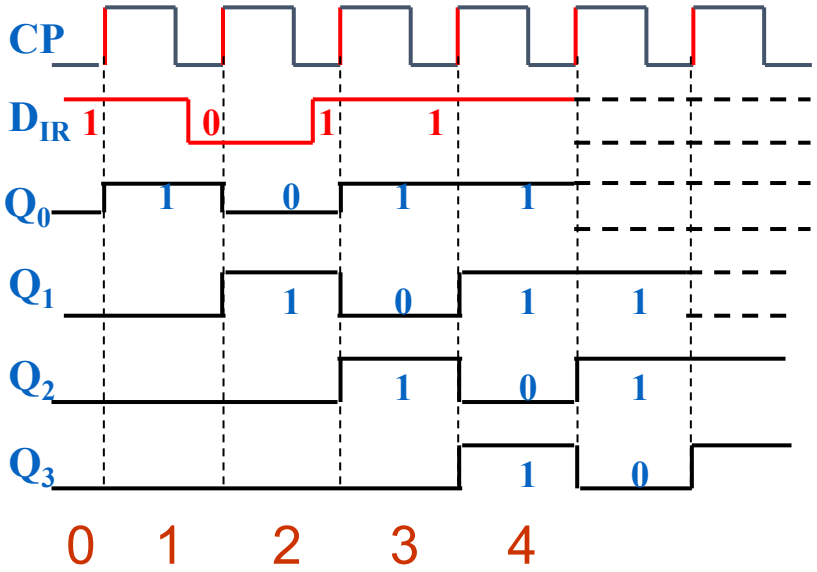
❖ 工作原理

◆ 复位：

$$\overline{R_D} = 0 \rightarrow Q_3Q_2Q_1Q_0 = 0000$$

◆ 移位：

$$Q_0^{n+1} = D_0\text{CP}\uparrow = D_{IR}\text{CP}\uparrow$$
$$Q_1^{n+1} = D_1\text{CP}\uparrow = Q_0^{n}\text{CP}\uparrow$$
$$Q_2^{n+1} = D_2\text{CP}\uparrow = Q_1^{n}\text{CP}\uparrow$$
$$Q_3^{n+1} = D_3\text{CP}\uparrow = Q_2^{n}\text{CP}\uparrow$$



串行输入、串/并行输出单向移位寄存器（2）

❖ 工作方式

- 串入并出

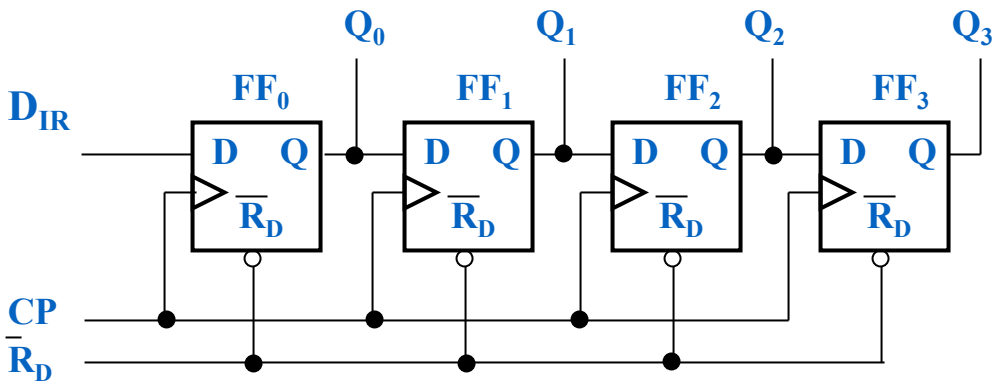
  串并转换（需要N个CP周期），

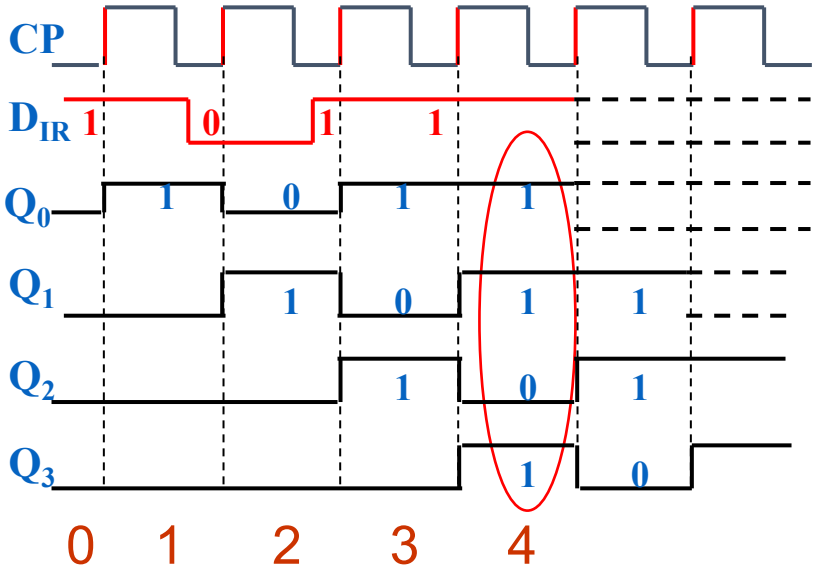
  经过4个CP，串行输入的4位数码全部移入移位寄存器中，并从$Q_3Q_2Q_1Q_0$并行输出1011

- 串入串出

  最右边的触发器的输出作为电路的输出。经过4个CP后， Q3 输出的是最先串行输入的数码。

  从每个触发器Q端输出的波形相同，但后级触发器Q端输出波形比前级触发器Q端输出波形滞后一个时钟周期（第N级FF延迟N个CP周期）



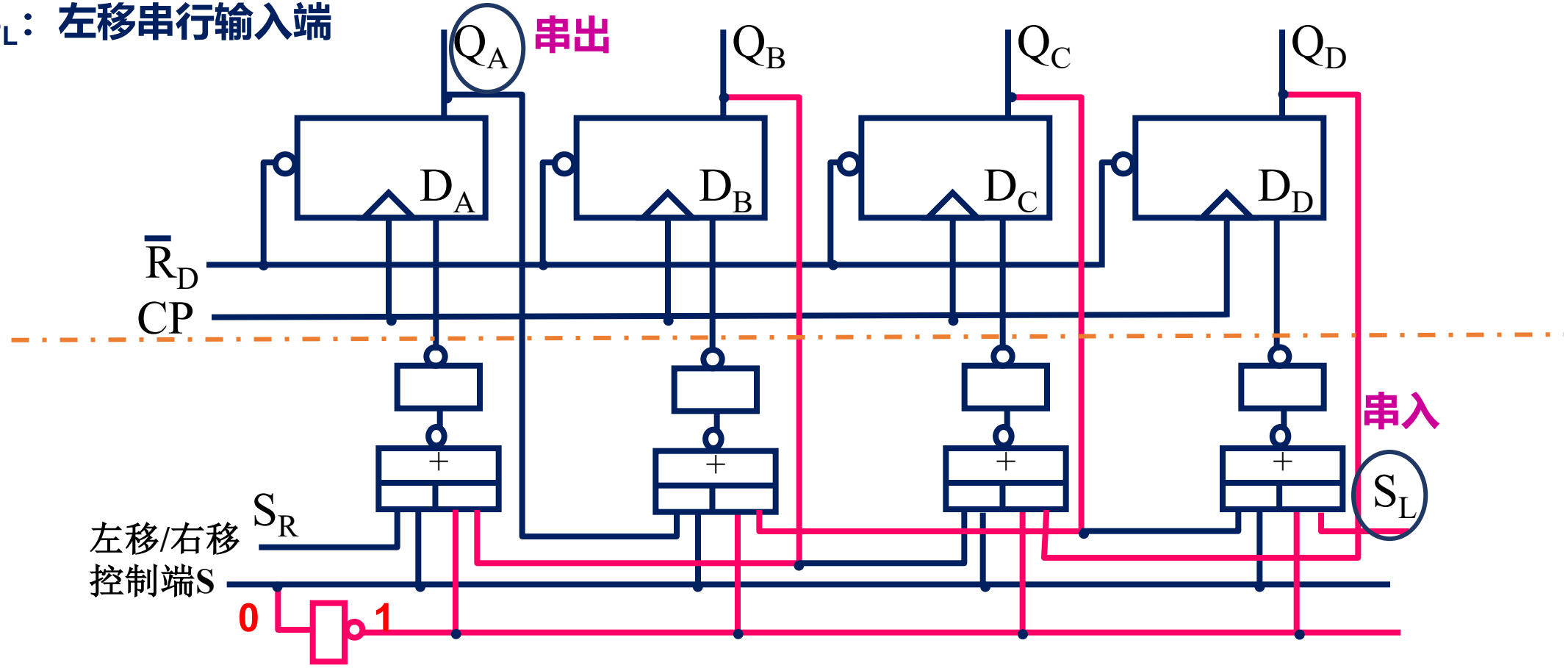


串行输入、串/并行输出双向移位寄存器

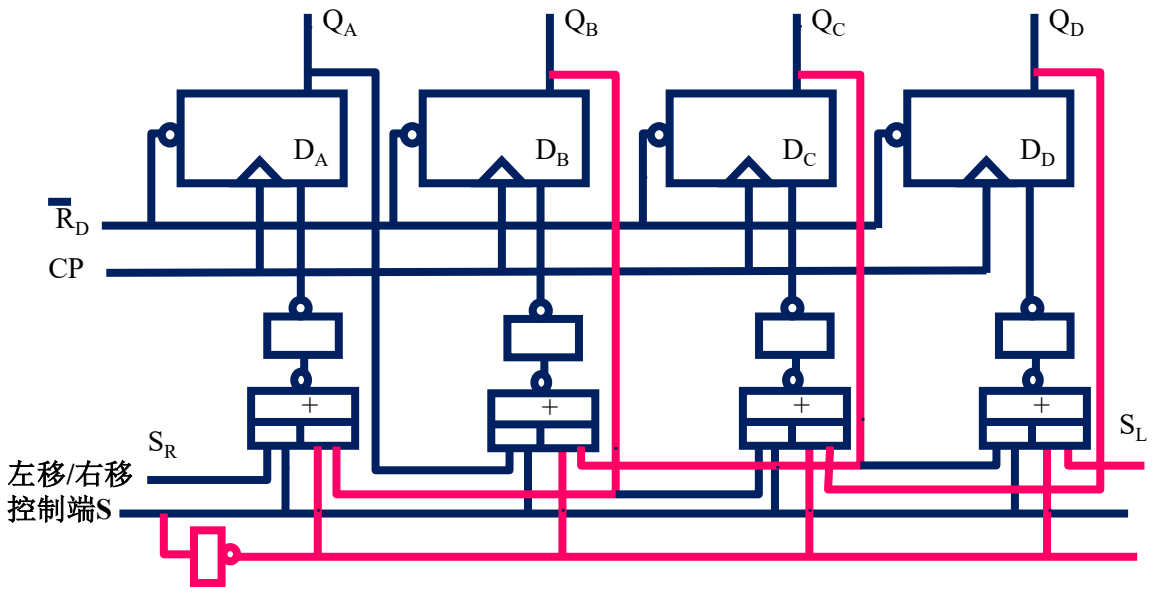
$D_A = S_R S + Q_B \bar{S}$　　$D_B = Q_A S + Q_C \bar{S}$

$D_C = Q_B S + Q_D \bar{S}$　　$D_D = Q_C S + S_L \bar{S}$

S：左移/右移控制端
S_R：右移串行输入端
S_L：左移串行输入端



串行输入、串/并行输出双向移位寄存器



- 复位 $\overline{R_D} = 0 \rightarrow Q_A Q_B Q_C Q_D = 0000$

- 移位

（1）当S=0 时：左移移位寄存器

串入并出——数据从S_L端串行输入，顺序左移，经过4个CP，串行输入的4位数码全部移入移位寄存器中，并从$Q_A Q_B Q_C Q_D$并行输出。

串入串出——最左边触发器的输出作为电路的输出。左移时，经过4个CP，最先串行输入的1位数码从Q_A输出，下一个CP上升沿到来时，Q_A输出串行移入的第2个数码……。

（2）S＝1时：右移移位寄存器

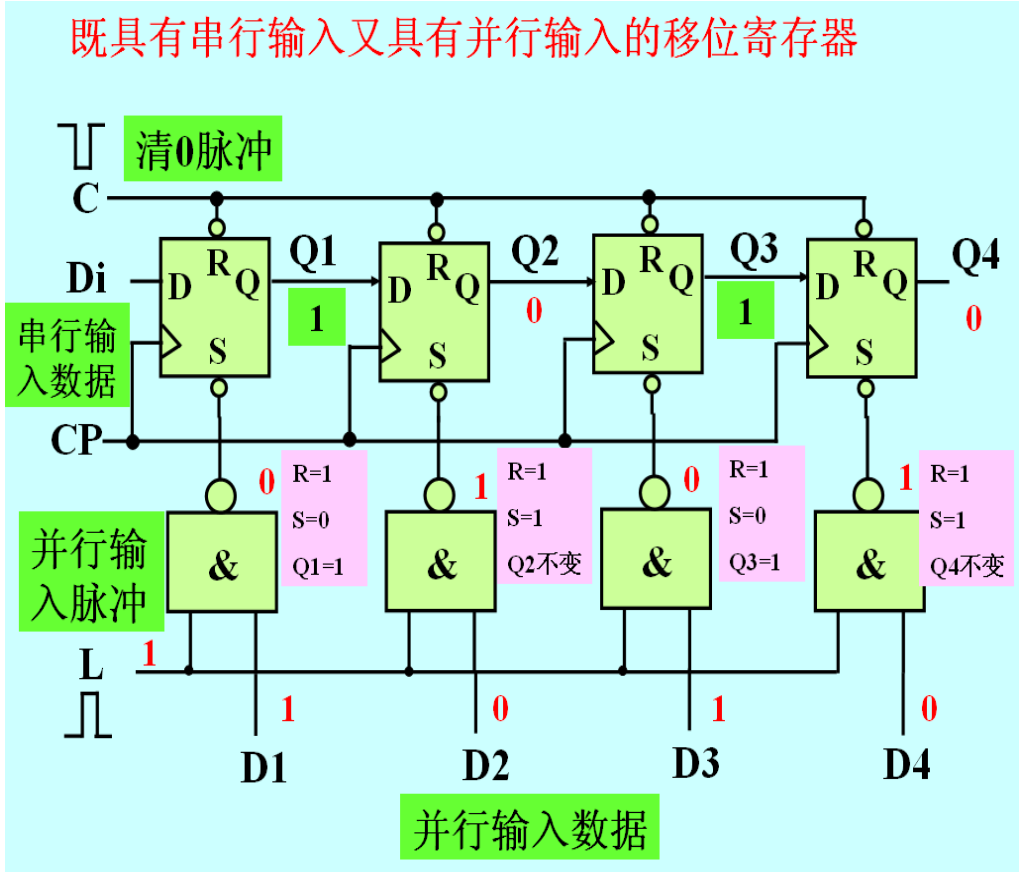
数据从S_R端串行输入，顺序右移。

集成移位寄存器

按输入、输出方式，集成移位寄存器分为5类

- 串入—并出单向移位寄存器
- 串入—串出单向移位寄存器
- 串入、并入—串出单向移位寄存器
- 串入、并入—并出单向移位寄存器
- 串入、并入—并出双向移位寄存器

串入、并入—串出单向移位寄存器

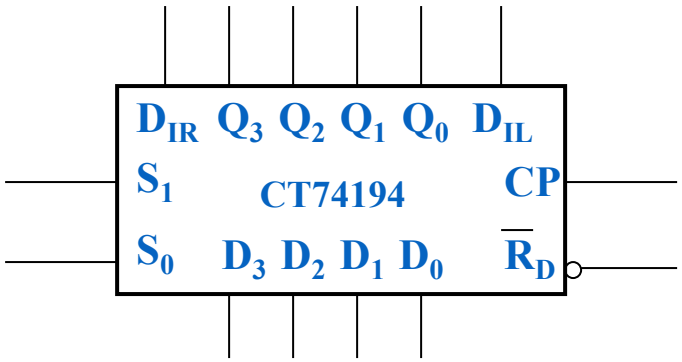


当C=0时DFF清0， $Q_1Q_2Q_3Q_4=0000$

当L为正脉冲、$D_1D_2D_3D_4=1010$时，

- 并行输入数据， 4个与非门的输出从左至右依次为0101；
- $Q_1Q_2Q_3Q_4=1010$；
- 经过4个CP后，并行输入的4位数据从Q_4 串行移出(0101)，最后输出的数据是D_1。

串入、并入—串出双向移位寄存器CT74194



S_1S_0：功能控制输入端

$D_3D_2D_1D_0$：并行数据输入端

$Q_3Q_2Q_1Q_0$：数据输出

D_{IR}：右移串行输入（Q_3为串行输出端）

D_{IL}：左移串行输入（Q_0为串行输出端）

功能表

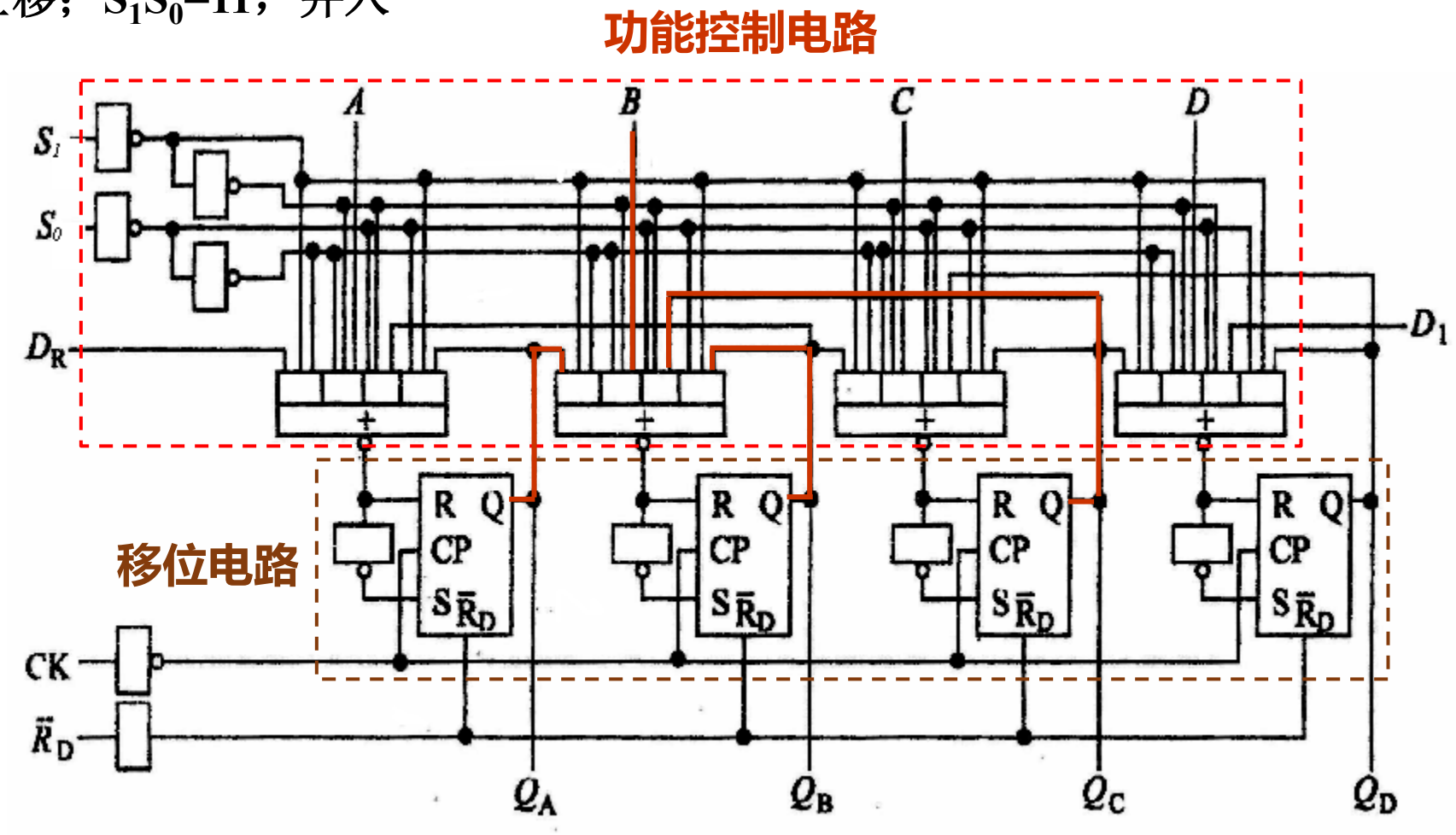
$\overline{R}_D$	CP	S_1	S_0	功能
0	X	X	X	置零
1	↑	0	0	保持
1	↑	0	1	右移
1	↑	1	0	左移
1	↑	1	1	并行输入

工作方式

- 串入并出——串并转换
- 串入串出——延迟线
- 并入串出——并串转换
- 并入并出——数据预置

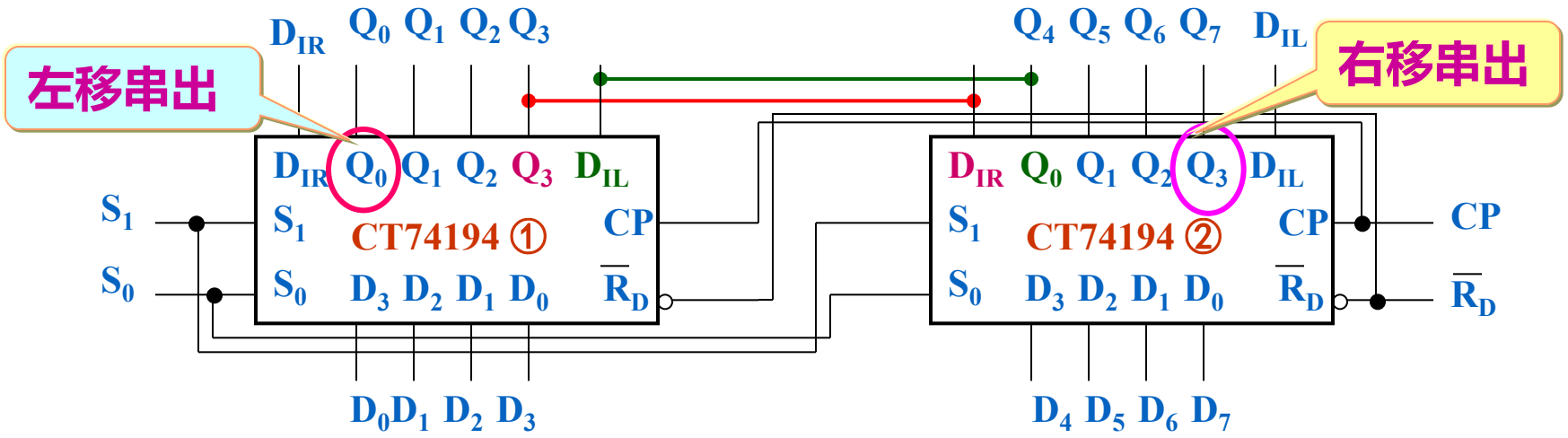
CT74194电路结构图

S_1S_0=00，保持；S_1S_0=01，右移；
S_1S_0=10，左移；S_1S_0=11，并入



集成移位寄存器扩展方法

用2片4位移位寄存器扩展为8位移位寄存器



- 将片①的Q_3接至片②的D_{IR}，当S1S0=01时，右移，片②的输出Q_3作为整个电路的右移串行输出端（Q_7）。
- 将片②的Q_0接至片① 的D_{IL}，当S1S0=10时，左移，片①的输出Q_0 作为整个电路的左移串行输出端Q_0。
- 同时把两片的S_1、S_0、CP和/R_D分别并联。

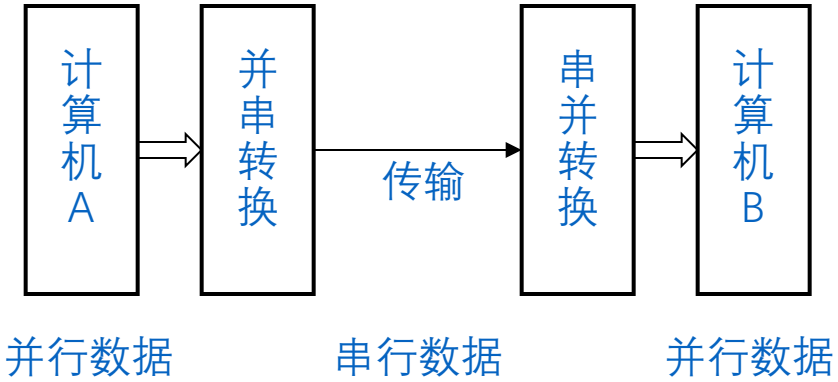
集成移位寄存器的用途

- 主要用途
  - 数据保存与移位
  - 计算机串行通信（使用串行接口RS232、RS485）中的并串转换及串并转换

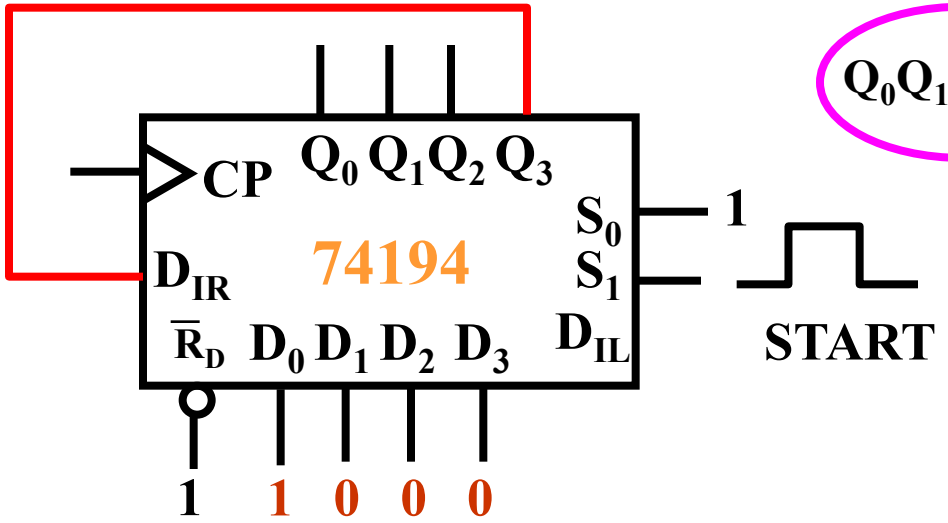
    串行通信——数据在一根传输线上一位一位地顺序传送。

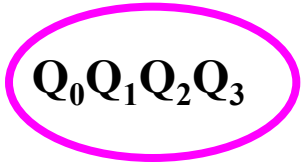
    并行通信——数据以字节（字）为单位在多根传输线上同时传送。
  - 移存型计数器（环形计数器，扭环形计数器）

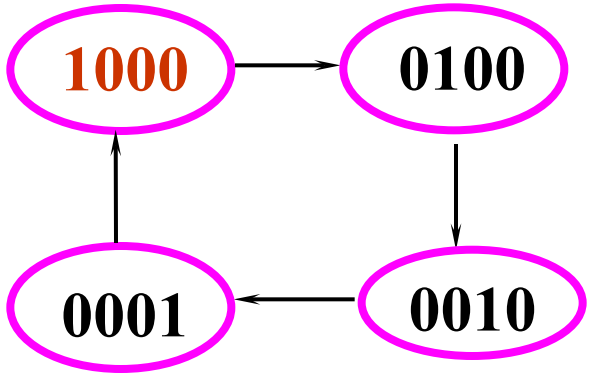
    利用移位寄存器组成的计数器叫做移存型计数器 。



用移位寄存器实现环形计数器

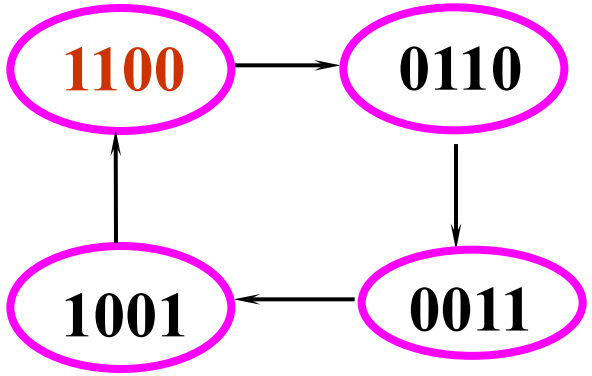


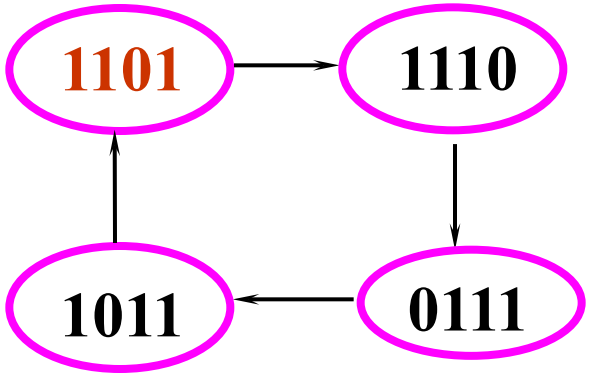


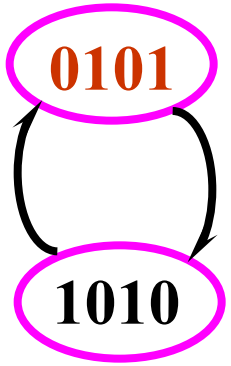


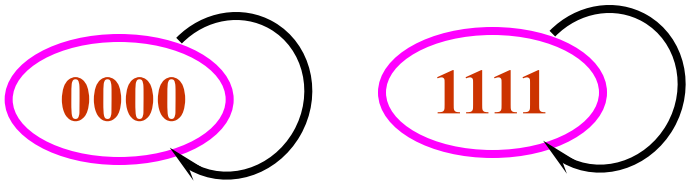
- 当$D_0D_1D_2D_3$为不同的初值，环形计数器的计数规律不同

- 当START信号脉冲有效时， S_1S_0=11，并行输入。
- 当START信号脉冲无效时， S_1S_0=01，执行右移。



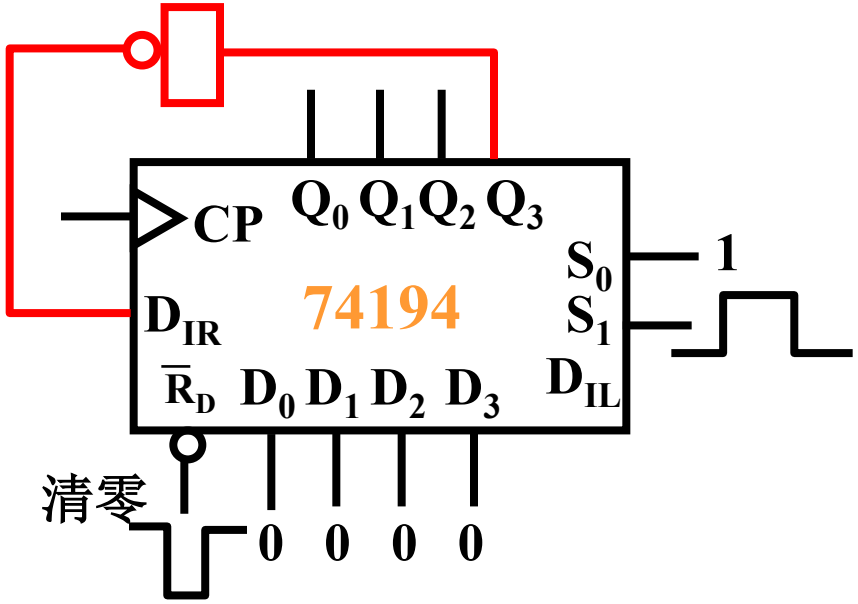


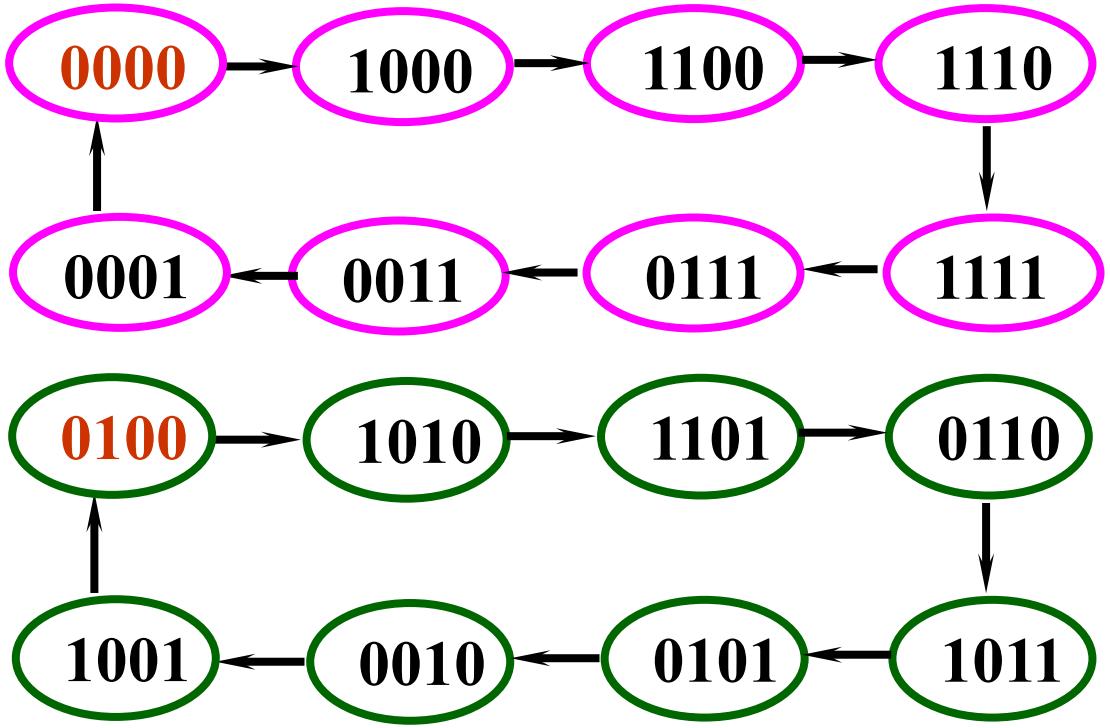




用移位寄存器实现扭环形计数器

- Q_3取反后接至右串行输入D_{IR}端。
- 当 START 信号脉冲有效时，S_1S_0=11，并行输入。
- 当 START 信号脉冲无效时，S_1S_0=01，执行右移。

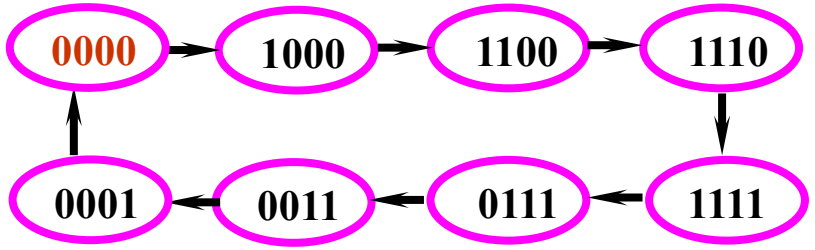


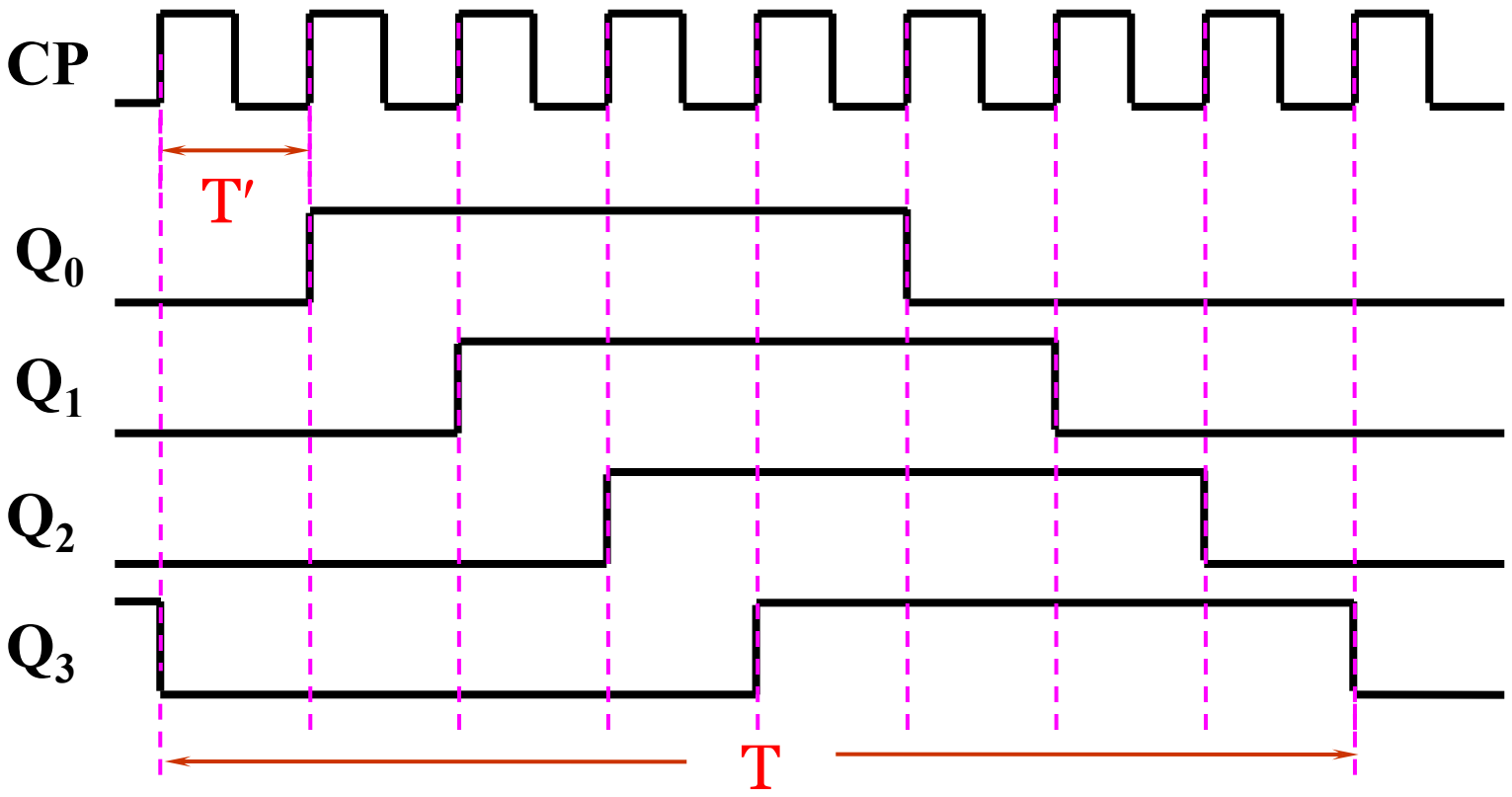


4位格雷码计数器（模8）

普通八进制计数器

4位格雷码计数器输出波形

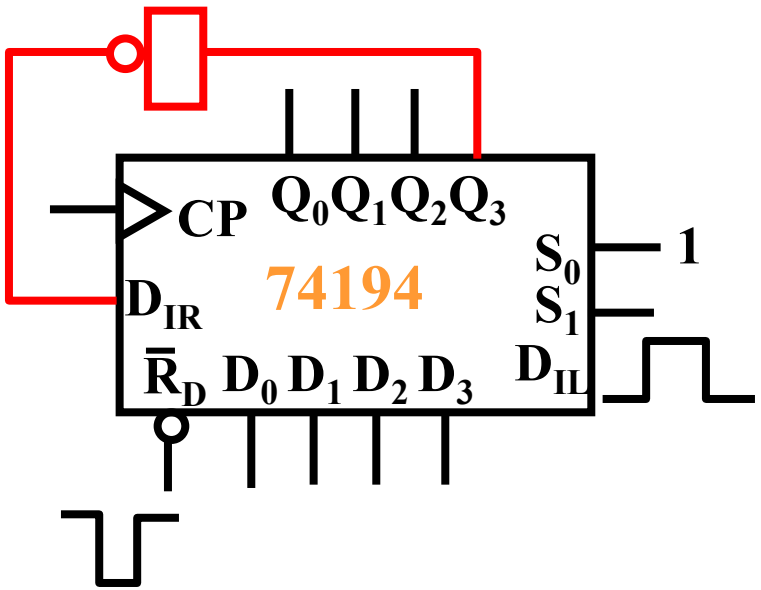




因为输出信号的频率为CP的1/8，所以4位格雷码计数器相当于÷8除法器

4位移位寄存器的应用——节日彩灯控制电路

【例10.1】利用4位双向移位寄存器CT74194设计一个节日彩灯控制电路，使其控制8个发光二极管，当按下清0键时，8个LED都亮；然后从第1个LED开始，每来一个CP，各LED依次熄灭；当第8个LED熄灭后，从第1个LED开始，每来一个CP， 各LED又依次点亮；重复此规律。



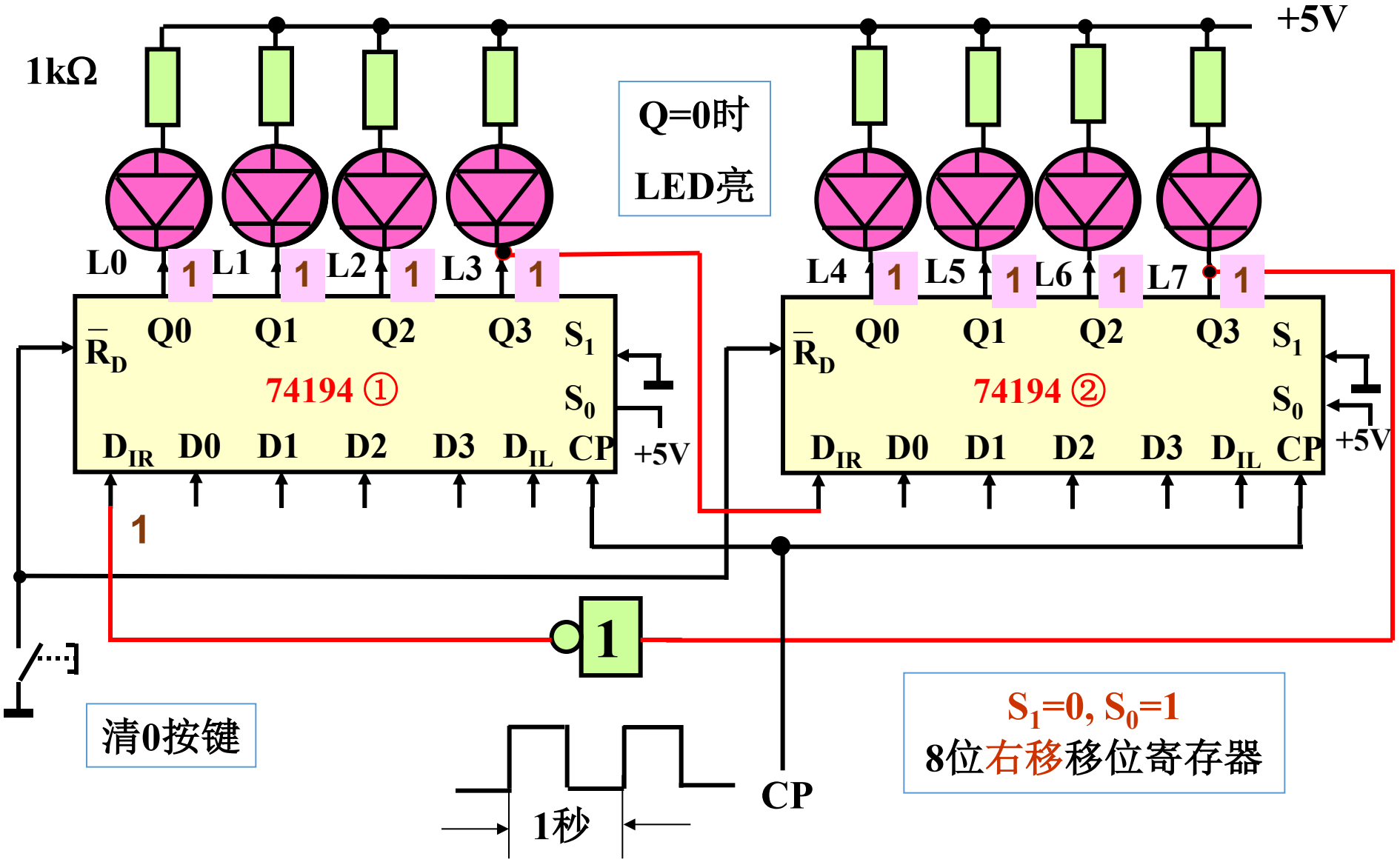
Q_0	Q_1	Q_2	Q_3
0	0	0	0
1	0	0	0
1	1	0	0
1	1	1	0
1	1	1	1
0	1	1	1
0	0	1	1
0	0	0	1

4位移位寄存器的应用——节日彩灯控制电路

【例10.1】利用4位双向移位寄存器CT74194设计一个节日彩灯控制电路，使其控制8个发光二极管，当按下清0键时，8个LED都亮；然后从第1个LED开始，每来一个CP，各LED依次熄灭；当第8个LED熄灭后，从第1个LED开始，每来一个CP， 各LED又依次点亮；重复此规律。

- 因为要控制8个发光二极管，所以需2片CT74194。
- 由于要求按下清0键时，8个LED都亮，则CT74194的的各输出端应接LED 的阴极。
- 利用CT74194的右移功能，Q3为串行输出端；将片①的Q_3接至片②的D_{IR}，构成8位右移移位寄存器；要构成扭环形计数器，故将片②的Q_3反相后接至片① 的D_{IR} 。

节日彩灯控制电路连接图



节日彩灯控制电路状态转移表

① Q_0	Q_1	Q_2	Q_3	② Q_0	Q_1	Q_2	Q_3
0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0
1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	0
1	1	1	1	1	0	0	0
1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1
0	0	1	1	1	1	1	1
0	0	0	1	1	1	1	1
0	0	0	0	1	1	1	1
0	0	0	0	0	1	1	1
0	0	0	0	0	0	1	1
0	0	0	0	0	0	0	1

- 把两片4位移位寄存器的CP和/R_D分别并联； S_1S_0=01，移位寄存器执行右移功能。
- 首先按清0键，移位寄存器清零，①和②的输出$Q_0Q_1Q_2Q_3$均为0，则8个LED都亮。
- 由于片①的Q_3接至片②的D_{IR}，则构成8位右移移位寄存器。
- 由于将片②的Q_3反相后接至片① 的D_{IR}，则在第1个CP作用下，片②的/Q_3=1进入片①的Q_0，使所驱动的L0熄灭，以后“1”不断移入Q_0，则其后的LED依次熄灭；
- 在第8个CP来到后，片①的D_{IR}=片②的/Q_3=0，以后“0”不断移入片①的Q_0。则从L0开始，每来一个CP，LED依次点亮。

10.2 计数器

1. 同步计数器
2. 异步计数器
3. 集成计数器

- 计数器是可以统计输入脉冲个数的器件
- 计数器的用途
  - （脉冲）计数
  - 定时（定时器）
  - 分频——将高频时钟信号分频为较低频率的信号
  - 产生节拍脉冲（顺序脉冲）
- 计数器的分类
  - 时钟：同步计数器，异步计数器
  - 计数方式：二进制计数器，十进制计数器，M进制计数器
  - 状态变化：加法计数器，减法计数器，加/减法计数器

一、同步计数器

1.1 同步二进制计数器

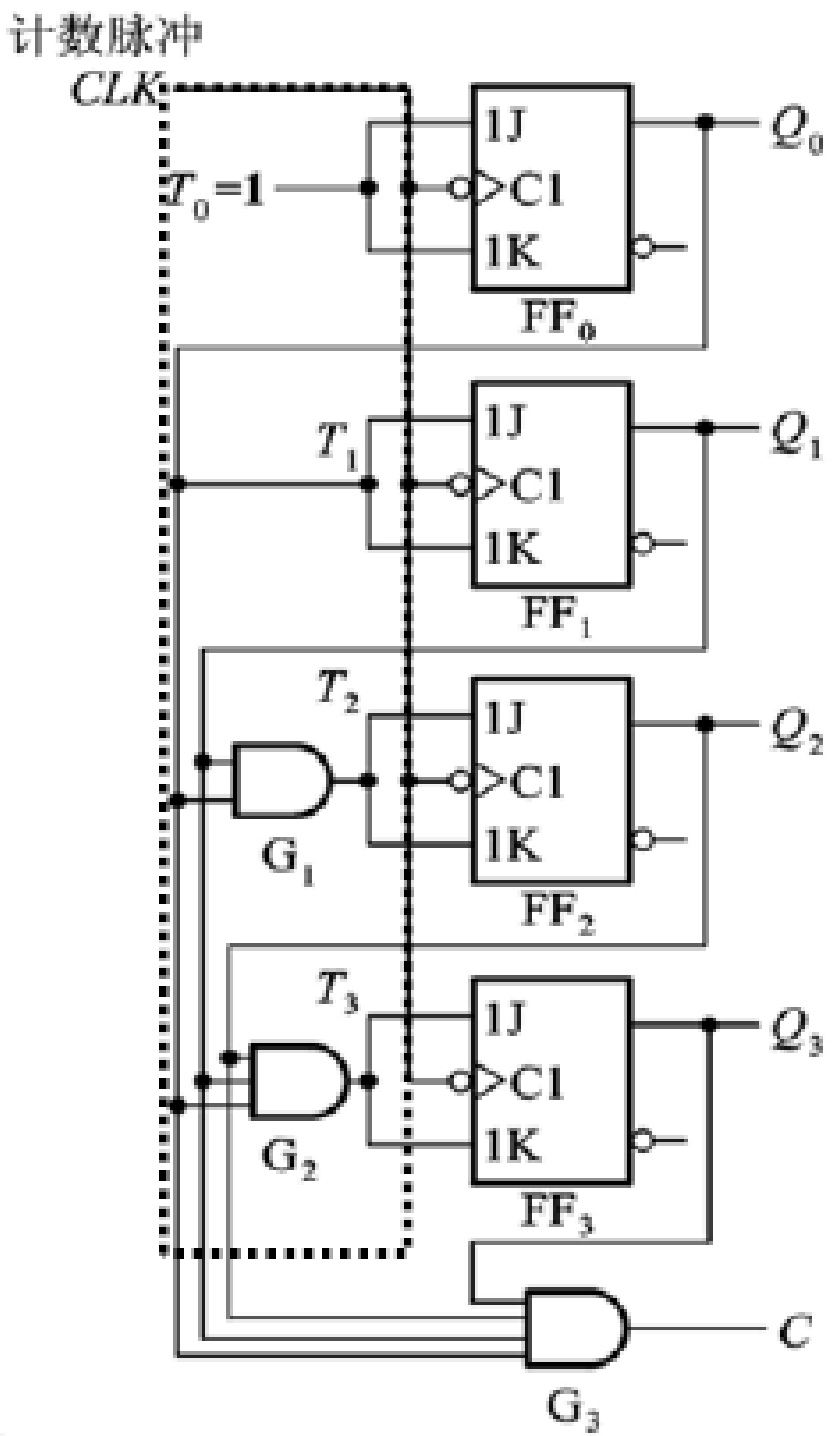
◆ 同步二进制加法计数器

原理：在多位二进制数末位加1，若第i位以下皆为1时，则第i位应翻转。

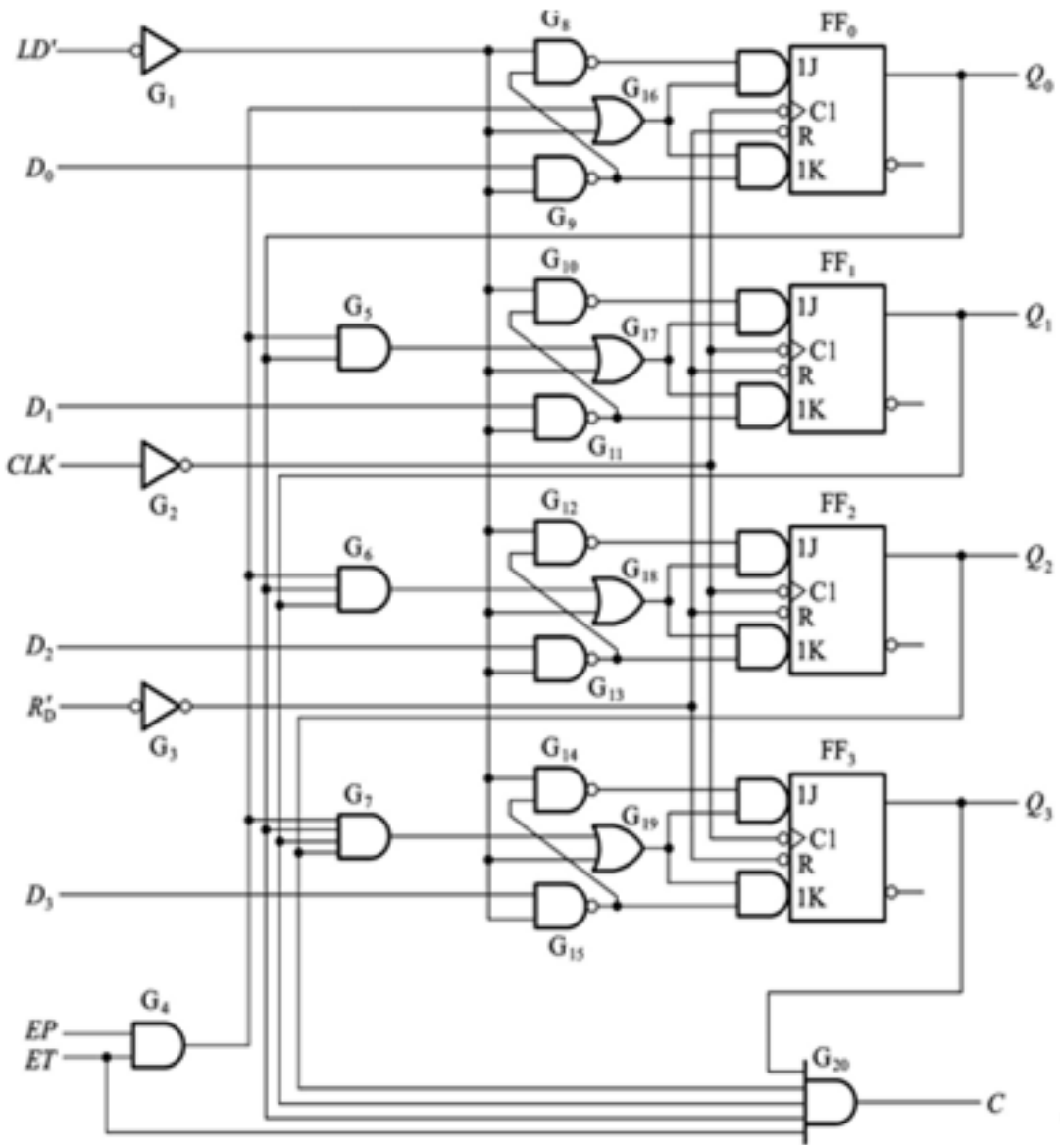
若用***T***触发器构成计数器，则第***i***位触发器输入端T_i的逻辑式应为：

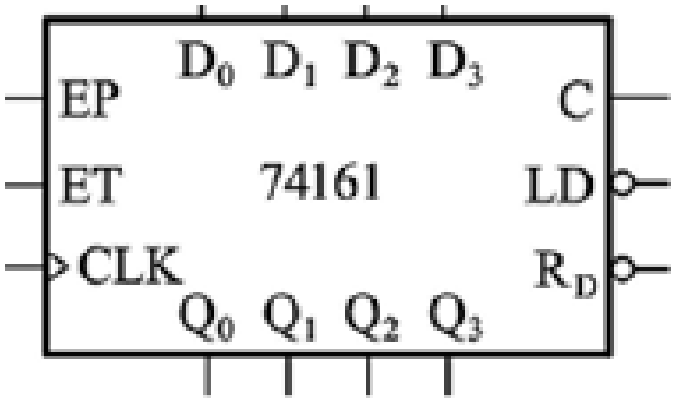
$$T_i = Q_{i-1}Q_{i-2}\cdots Q_0$$

$$T_0 \equiv 1$$



器件实例 74161





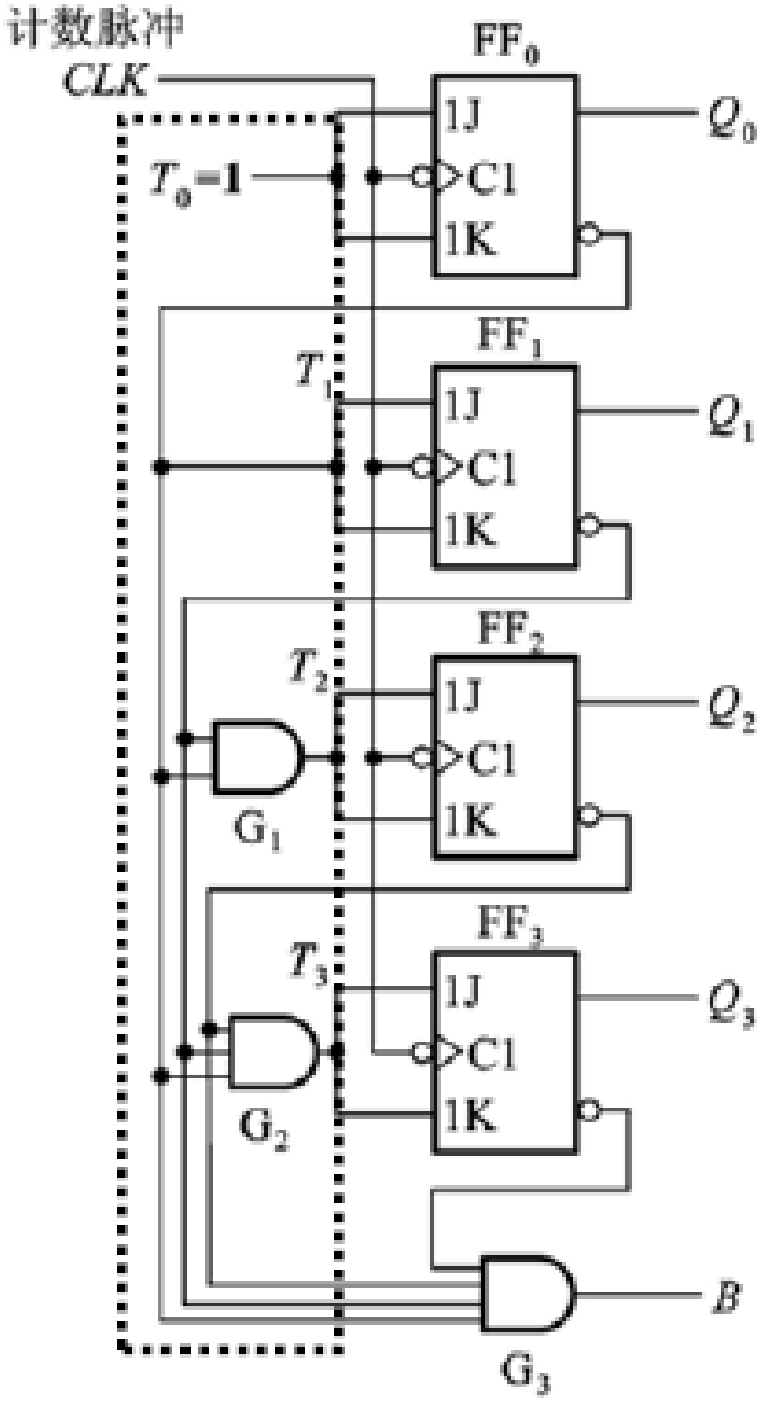
CLK	R'_D	LD'	EP	ET	工作状态
X	0	X	X	X	置 0（异步）
↑	1	0	X	X	预置数（同步）
X	1	1	0	1	保持（包括C）
X	1	1	X	0	保持（C=0）
↑	1	1	1	1	计数

◆ 同步二进制减法计数器

原理： 在多位二进制数末位减1，若第i位以下皆为0时，则第i位应翻转。

若用***T***触发器构成计数器，则第***i***位触发器输入端T_i的逻辑式应为：

$$T_i = Q'_{i-1}Q'_{i-2}\cdots Q'_0$$
$$T_0 \equiv 1$$



◆ 同步加减计数器

a. 单时钟方式

加/减脉冲用同一输入端，

由加/减控制线的高低电平决定加/减

$$\begin{cases} T_i = (U'/D')\prod_{j=0}^{i-1} Q_j + (U'/D')\prod_{j=0}^{i-1} Q'_j \\ T_0 = 1 \end{cases}$$

器件实例　74191

b.双时钟方式

器件实例　74193

FAIRCHILD SEMICONDUCTOR™

August 1986
Revised February 1999

DM74LS191
Synchronous 4-Bit Up/Down Counter with Mode Control

General Description

The DM74LS191 circuit is a synchronous, reversible, up/down counter. Synchronous operation is provided by having all flip-flops clocked simultaneously, so that the outputs change simultaneously when so instructed by the steering logic. This mode of operation eliminates the output counting spikes normally associated with asynchronous (ripple clock) counters.

The outputs of the four master-slave flip-flops are triggered on a LOW-to-HIGH level transition of the clock input, if the enable input is LOW. A HIGH at the enable input inhibits counting. Level changes at either the enable input or the down/up input should be made only when the clock input is HIGH. The direction of the count is determined by the level of the down/up input. When LOW, the counter counts up and when HIGH, it counts down.

The counter is fully programmable; that is, the outputs may be preset to either level by placing a LOW on the load input and entering the desired data at the data inputs. The output will change independent of the level of the clock input. This feature allows the counters to be used as modulo-N dividers by simply modifying the count length with the preset inputs.

The clock, down/up, and load inputs are buffered to lower the drive requirement; which significantly reduces the number of clock drivers, etc., required for long parallel words.

Two outputs have been made available to perform the cascading function: ripple clock and maximum/minimum count. The latter output produces a high-level output pulse with a duration approximately equal to one complete cycle of the clock when the counter overflows or underflows. The ripple clock output produces a low-level output pulse equal in width to the low-level portion of the clock input when an overflow or underflow condition exists. The counters can be easily cascaded by feeding the ripple clock output to the enable input of the succeeding counter if parallel clocking is used, or to the clock input if parallel enabling is used. The maximum/minimum count output can be used to accomplish look-ahead for high-speed operation.

Features

- Counts binary
- Single down/up count control line
- Count enable control input
- Ripple clock output for cascading
- Asynchronously presettable with load control
- Parallel outputs
- Cascadable for n-bit applications
- Average propagation delay 20 ns
- Typical clock frequency 25 MHz
- Typical power dissipation 100 mW

DM74LS191 Synchronous 4-Bit Up/Down Counter with Mode Control

https://pdf1.alldatasheet.com/datasheet-pdf/view/51049/FAIRCHILD/74191.html

FAIRCHILD SEMICONDUCTOR™

September 1986
Revised March 2000

DM74LS193
Synchronous 4-Bit Binary Counter with Dual Clock

General Description

The DM74LS193 circuit is a synchronous up/down 4-bit binary counter. Synchronous operation is provided by having all flip-flops clocked simultaneously, so that the outputs change together when so instructed by the steering logic. This mode of operation eliminates the output counting spikes normally associated with asynchronous (ripple-clock) counters.

The outputs of the four master-slave flip-flops are triggered by a LOW-to-HIGH level transition of either count (clock) input. The direction of counting is determined by which count input is pulsed while the other count input is held HIGH.

The counter is fully programmable; that is, each output may be preset to either level by entering the desired data at the inputs while the load input is LOW. The output will change independently of the count pulses. This feature allows the counters to be used as modulo-N dividers by simply modifying the count length with the preset inputs.

A clear input has been provided which, when taken to a high level, forces all outputs to the low level; independent of the count and load inputs. The clear, count, and load inputs are buffered to lower the drive requirements of clock drivers, etc., required for long words.

These counters were designed to be cascaded without the need for external circuitry. Both borrow and carry outputs are available to cascade both the up and down counting functions. The borrow output produces a pulse equal in width to the count down input when the counter underflows. Similarly, the carry output produces a pulse equal in width to the count down input when an overflow condition exists. The counters can then be easily cascaded by feeding the borrow and carry outputs to the count down and count up inputs respectively of the succeeding counter.

Features

- Fully independent clear input
- Synchronous operation
- Cascading circuitry provided internally
- Individual preset each flip-flop

Ordering Code:

Order Number	Package Number	Package Description
DM74LS193M	M16A	16-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-012, 0.150" Narrow Body
DM74LS193N	N16E	16-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300" Wide

DM74LS193 Synchronous 4-Bit Binary Counter with Dual Clock

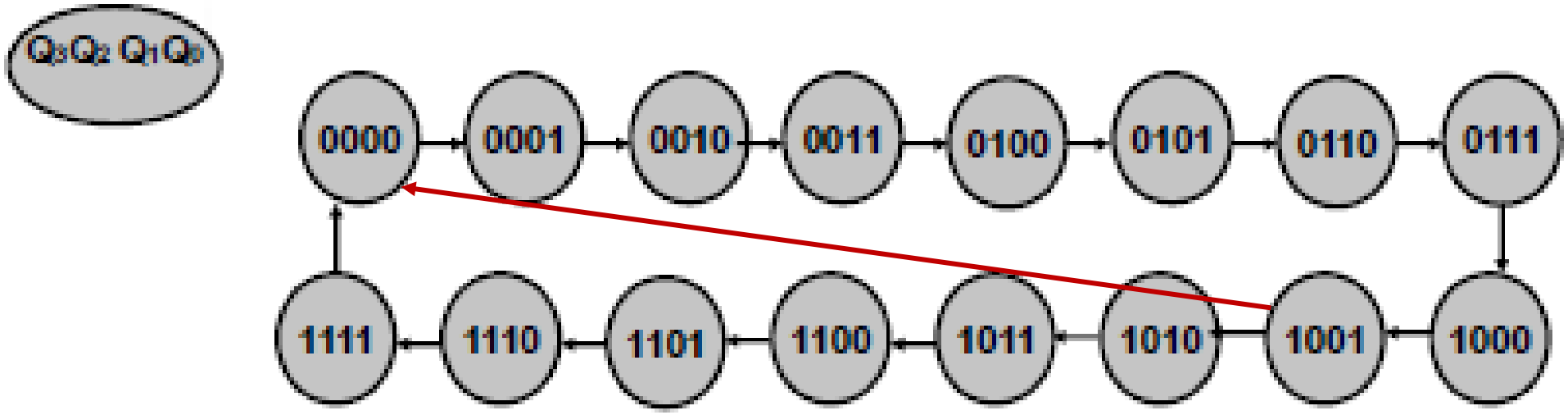
https://pdf1.alldatasheet.com/datasheet-pdf/view/51050/FAIRCHILD/74193.html

一、同步计数器

1.2 同步十进制计数器

◆ 加法计数器

原理： 在四位二进制计数器基础上修改，当计到**1001**时，则下一个**CLK**电路状态回到**0000**。

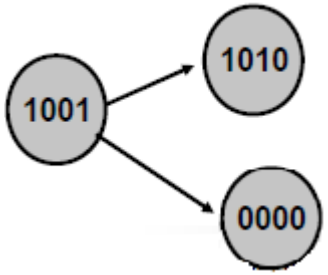


一、同步计数器

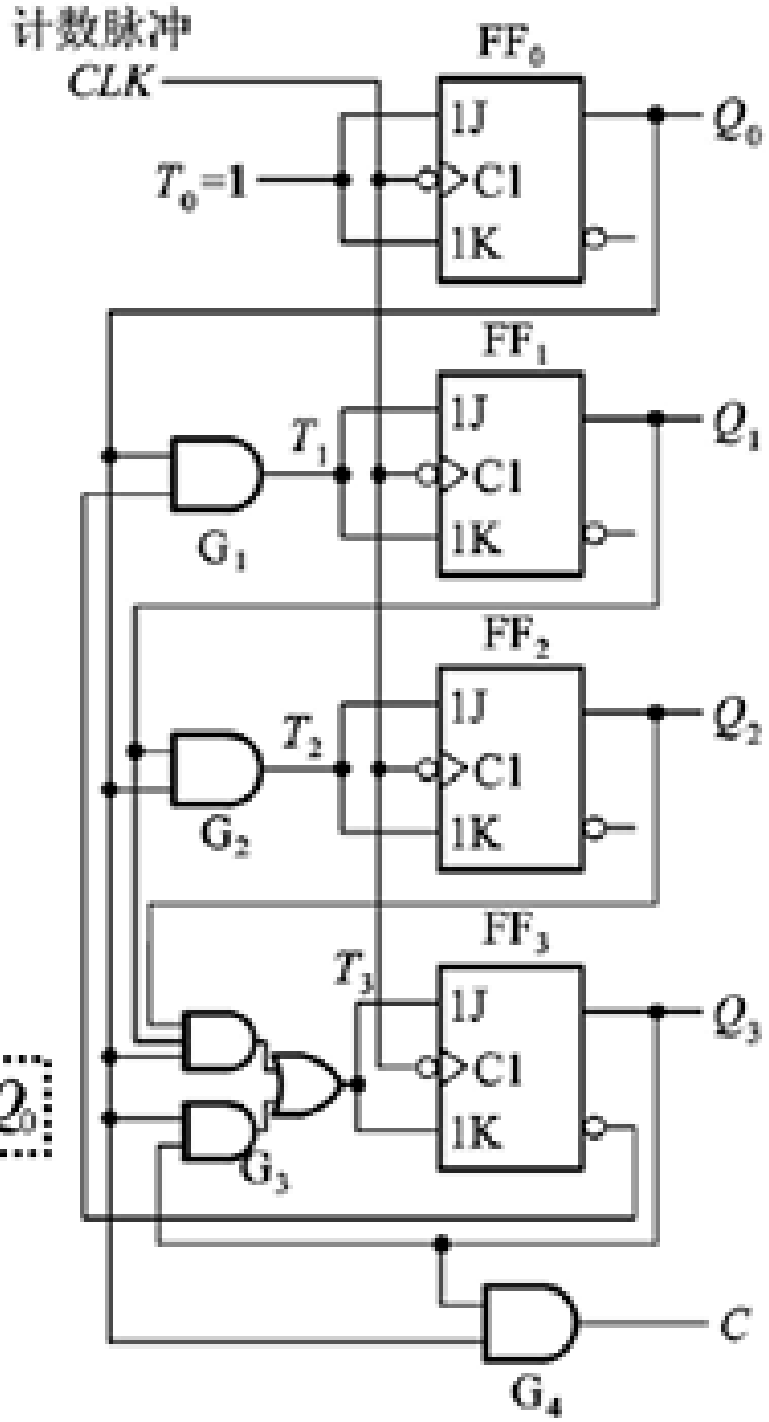
1.2 同步十进制计数器

◆ 加法计数器

基本原理：在四位二进制计数器基础上修改，当计到**1001**时，下一个**CLK**电路状态回到**0000**。



$$T_3 = Q_3Q_2Q_1Q_0 \Rightarrow Q_3Q_2Q_1Q_0 + Q_3Q_0$$

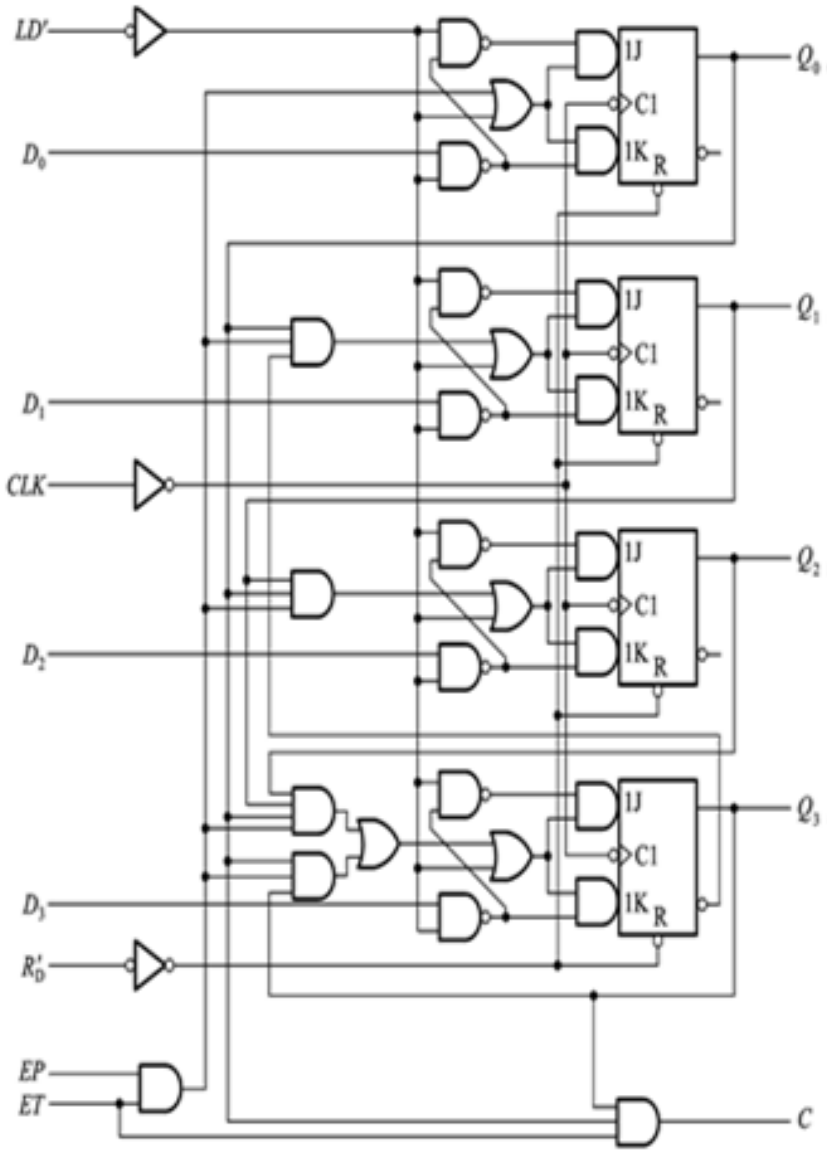


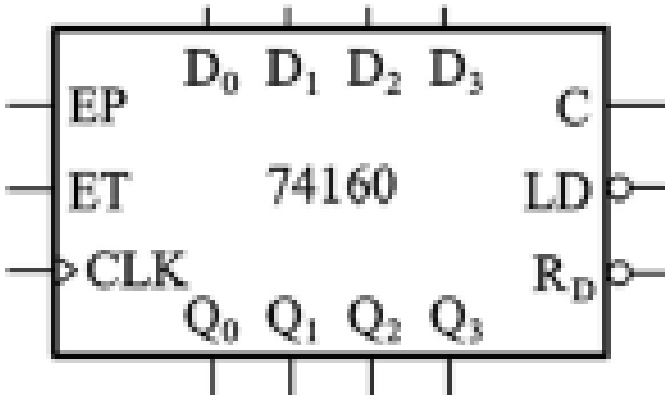
$$T_0 = 1$$

$$T_1 = Q_0 \Rightarrow Q_0Q_3$$

$$T_2 = Q_1Q_0$$

器件实例　74160

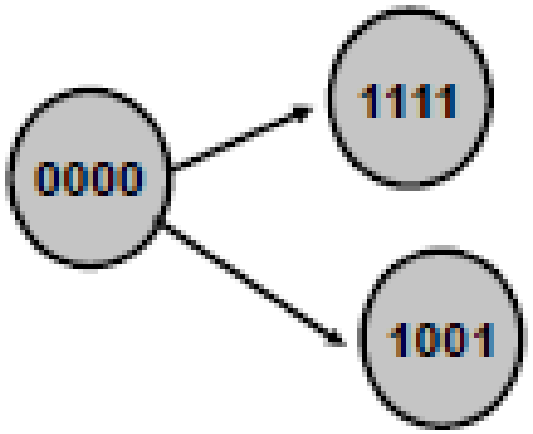


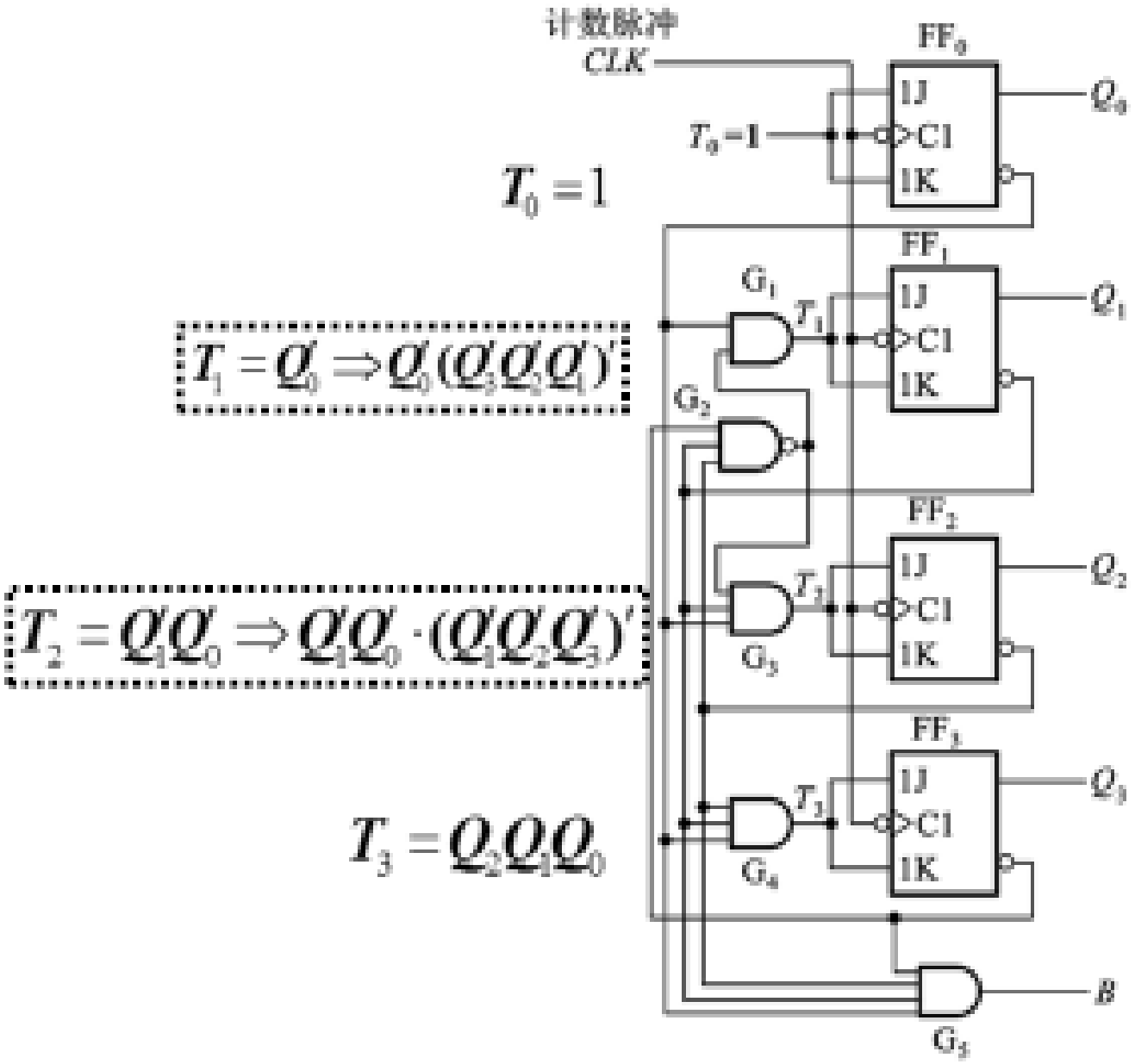


CLK	R'_D	LD'	EP	ET	工作状态
X	0	X	X	X	置 0（异步）
↑	1	0	X	X	预置数（同步）
X	1	1	0	1	保持（包括C）
X	1	1	X	0	保持（C=0）
↑	1	1	1	1	计数

◆ 减法计数器

基本原理：对二进制减法计数器进行修改，在**0000**时减“**1**”后跳变为**1001**，然后按二进制减法计数。





◆ 十进制可逆计数器

基本原理一致，电路只用到0000~1001的十个状态

器件实例

单时钟：74190,168

双时钟：74192

二、异步计数器

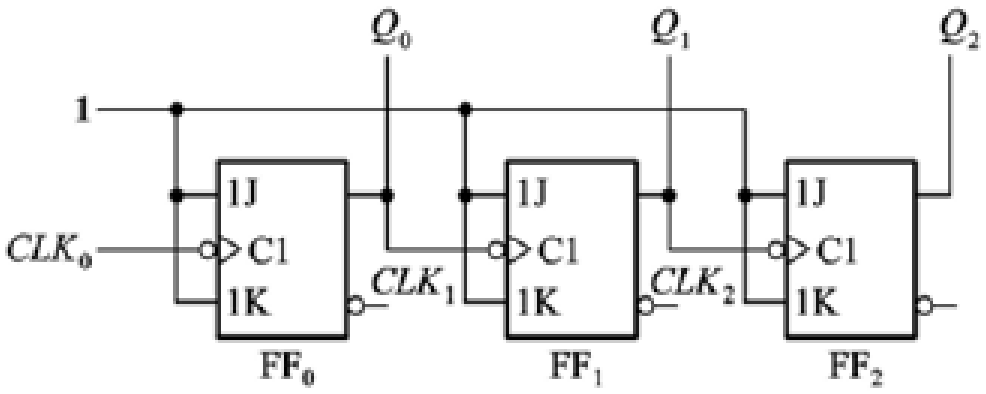
- 输入系统时钟脉冲只作用于最低位触发器，高位触发器的时钟信号往往是由低一位触发器的输出提供的，高位触发器的翻转有待低一位触发器翻转后才能进行。
- 由于每一级触发器都存在传输延迟，因此计数器工作速度慢，而且，位数越多计数越慢。在大型数字设备中较少采用。
- 对计数器状态进行译码时，由于触发器不同步，译码器输出会出现尖峰脉冲（位数越多，尖峰信号也就越宽），使仪器设备产生误动作。
- 优点：结构比较简单，所用元件较少。

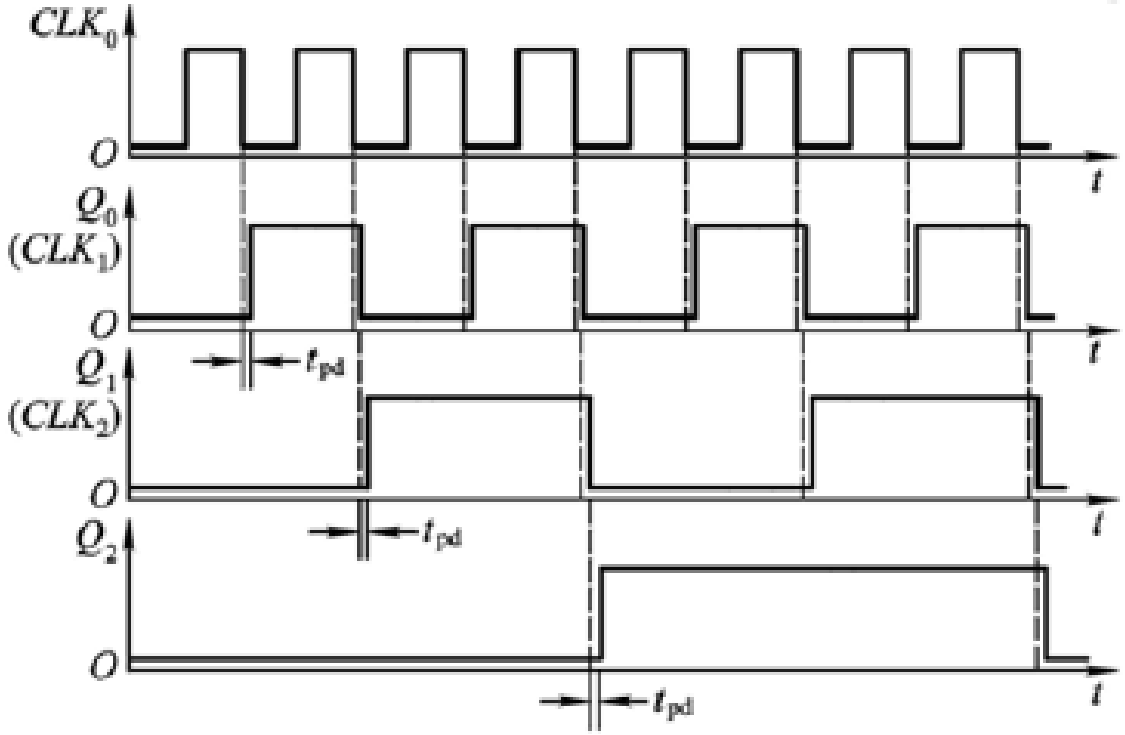
二、异步计数器

◆ 异步二进制加法计数器

在末位+1时，从低位到高位逐位进位方式工作。

原则：每1位从“1”变“0”时，向高位发出进位，使高位翻转。



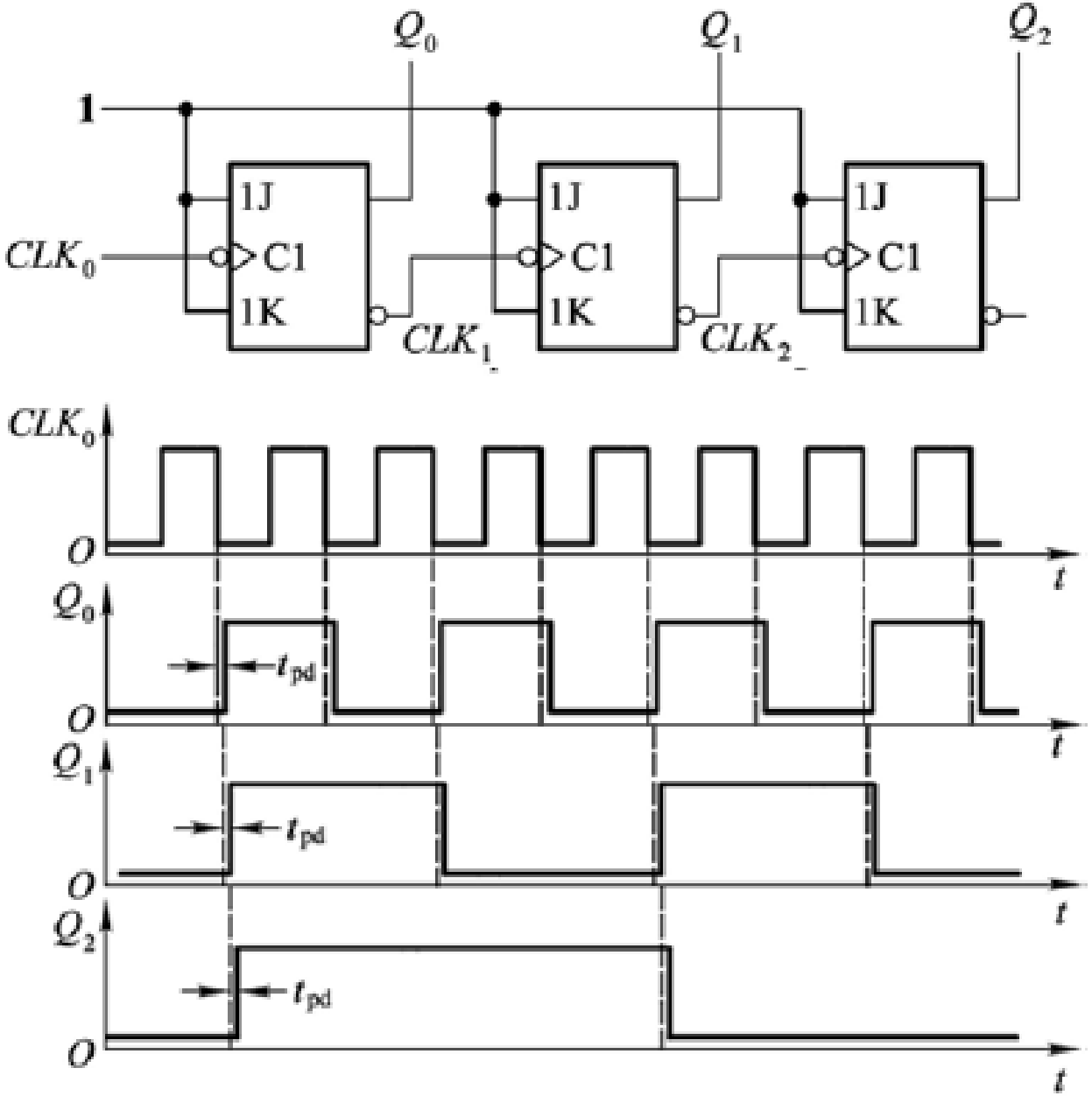


二、异步计数器

◆ 异步二进制减法计数器

在末位-1时，从低位到高位逐位借位方式工作。

原则：每1位从“0”变“1”时，向高位发出借位，使高位翻转。



n位二进制异步计数器总结

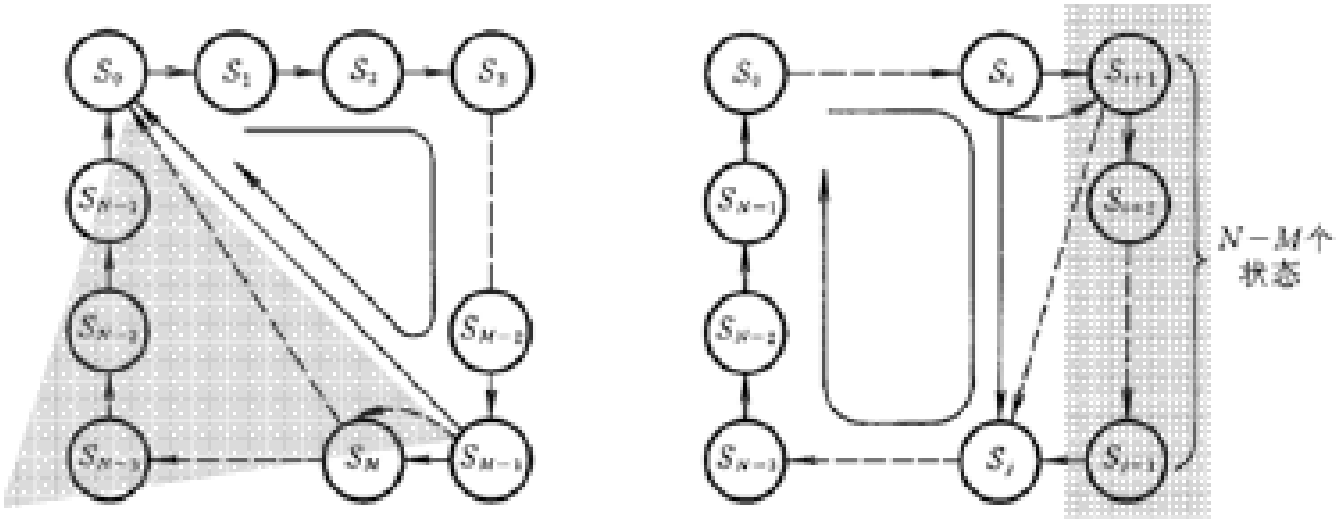
- n位二进制异步计数器由n个处于计数工作状态（对于D 触发器，使$D_i=/Q_i$；对于JK 触发器，使$J_i=K_i=1$）的触发器组成。各触发器之间的连接方式由加、减计数方式及触发器的触发方式决定。
  - 对于加计数器，若用上升沿触发的触发器组成，则应将低位触发器的Q非端与相邻高一位触发器的时钟脉冲输入端相连（即进位信号应从触发器的Q非端引出）；若下降沿触发，则应将低位触发器的Q端与相邻高一位触发器的时钟脉冲输入端连接。
  - 对于减计数器，各触发器的连接方式则相反。若触发器上升沿触发，则应将低位触发器的Q端与相邻高一位触发器的时钟脉冲输入端相连；若下降沿触发，则应将低位触发器的Q非端与相邻高一位触发器的时钟脉冲输入端连接。
- 在二进制异步计数器中，高位触发器的状态翻转必须在低一位触发器产生进位信号（加计数）或借位信号（减计数）之后才能实现。故又称这种类型的计数器为串行计数器。也正因为如此，异步计数器的工作速度较低。

三、任意进制计数器的构成方法

1. N > M

原理：计数循环过程中设法跳过N－M个状态。

具体方法：置零法　　置数法



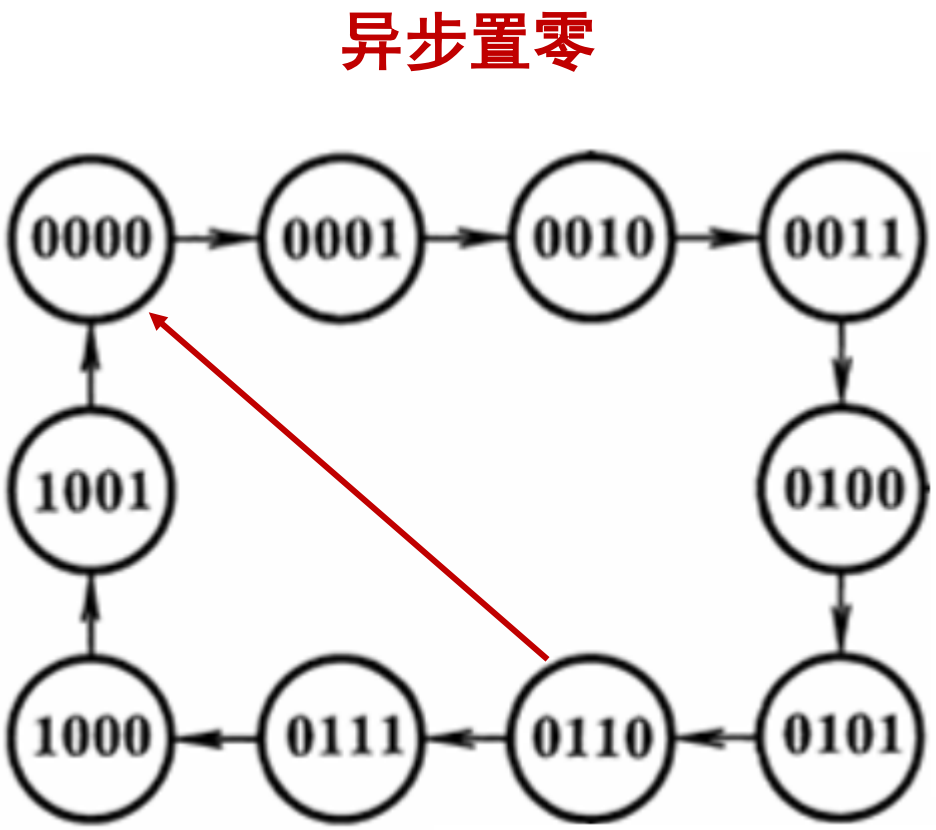
(a)　　(b)

$\begin{cases} \text{异步置零法} \\ \text{同步置零法} \end{cases}$　　$\begin{cases} \text{异步预置数法} \\ \text{同步预置数法} \end{cases}$

三、任意进制计数器的构成方法

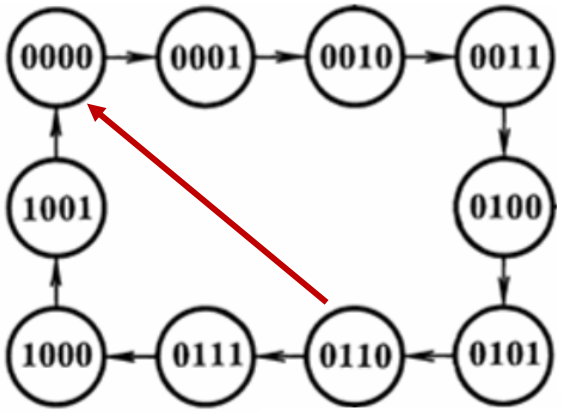
【例10.2】将十进制的74160接成六进制计数器。

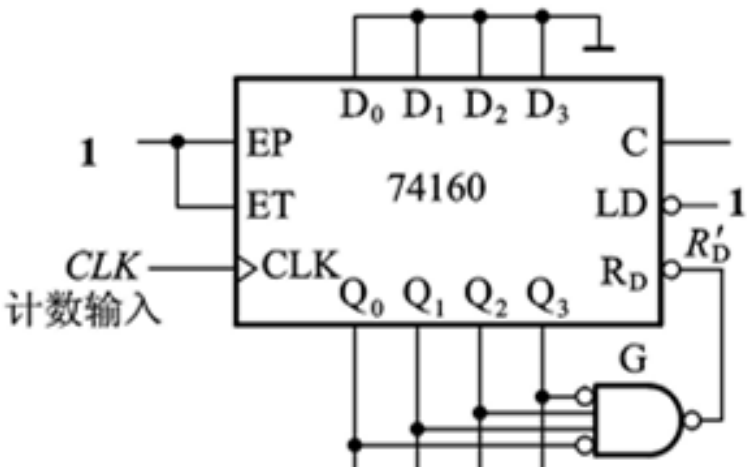
CLK	R'_D	LD'	EP	ET	工作状态
X	0	X	X	X	置 0（异步）
↑	1	0	X	X	预置数（同步）
X	1	1	0	1	保持（包括C）
X	1	1	X	0	保持（C=0）
↑	1	1	1	1	计数

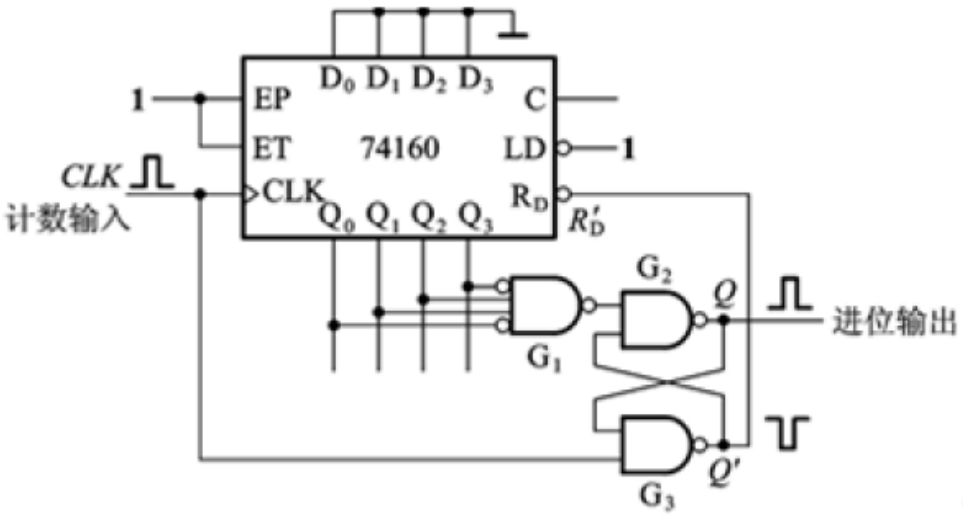


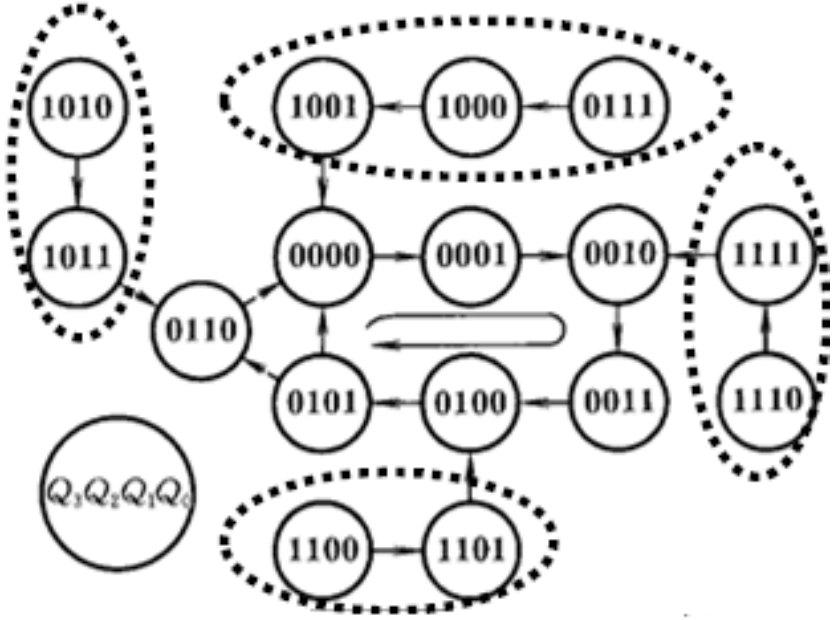
三、任意进制计数器的构成方法

【例10.2】将十进制的74160接成六进制计数器。





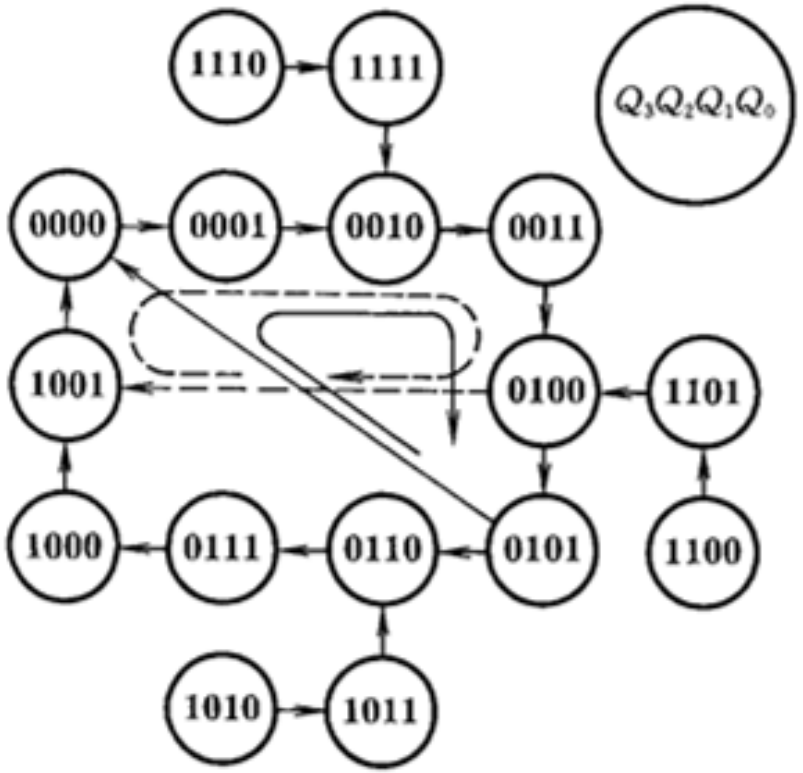


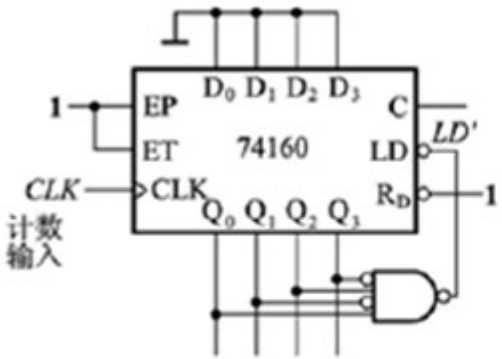


三、任意进制计数器的构成方法

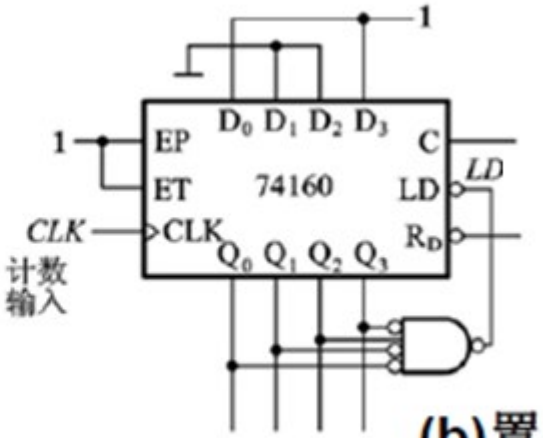
【例10.2】将十进制的74160接成六进制计数器。

CLK	R'_D	LD'	EP	ET	工作状态
X	0	X	X	X	置 0（异步）
↑	1	0	X	X	预置数（同步）
X	1	1	0	1	保持（包括C）
X	1	1	X	0	保持（C=0）
↑	1	1	1	1	计数





(a)置入0000



(b)置入1001

三、任意进制计数器的构成方法

2. N < M

①M=N1×N2

先用前面的方法分别接成N1和N2两个计数器。

N1和N2间的连接有两种方式：

a.并行进位方式：用同一个CLK，低位片的进位输出作为高位片的计数控制信号（如74160的EP和ET）

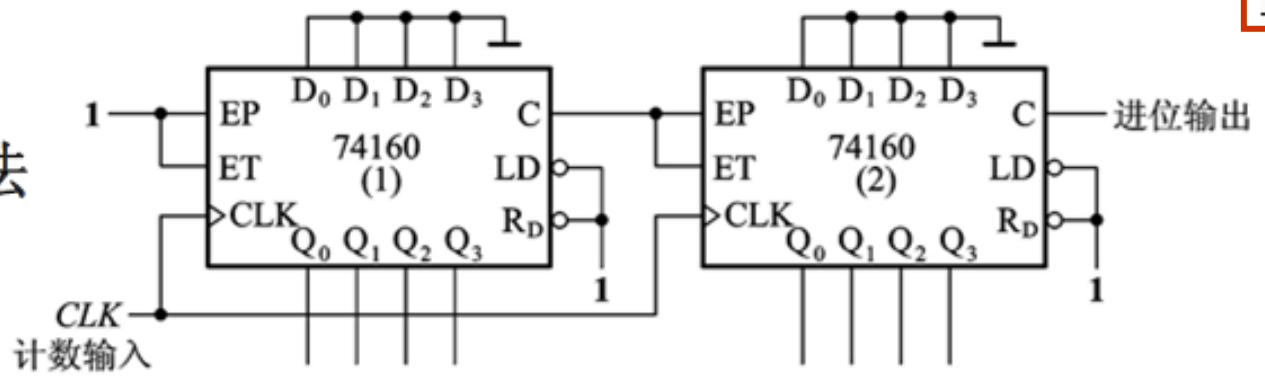
b.串行进位方式：低位片的进位输出作为高位片的CLK，两片始终同时处于计数状态

三、任意进制计数器的构成方法

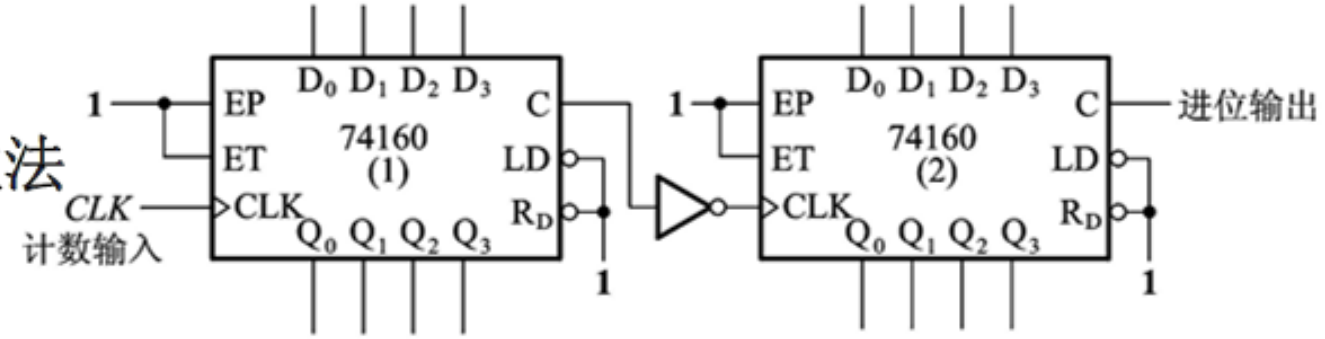
【例10.3】用两片74160接成100进制计数器。

CLK	R'_D	LD'	EP	ET	工作状态
X	0	X	X	X	置 0（异步）
↑	1	0	X	X	预置数（同步）
X	1	1	0	1	保持（包括C）
X	1	1	X	0	保持（C=0）
↑	1	1	1	1	计数

并行进位法

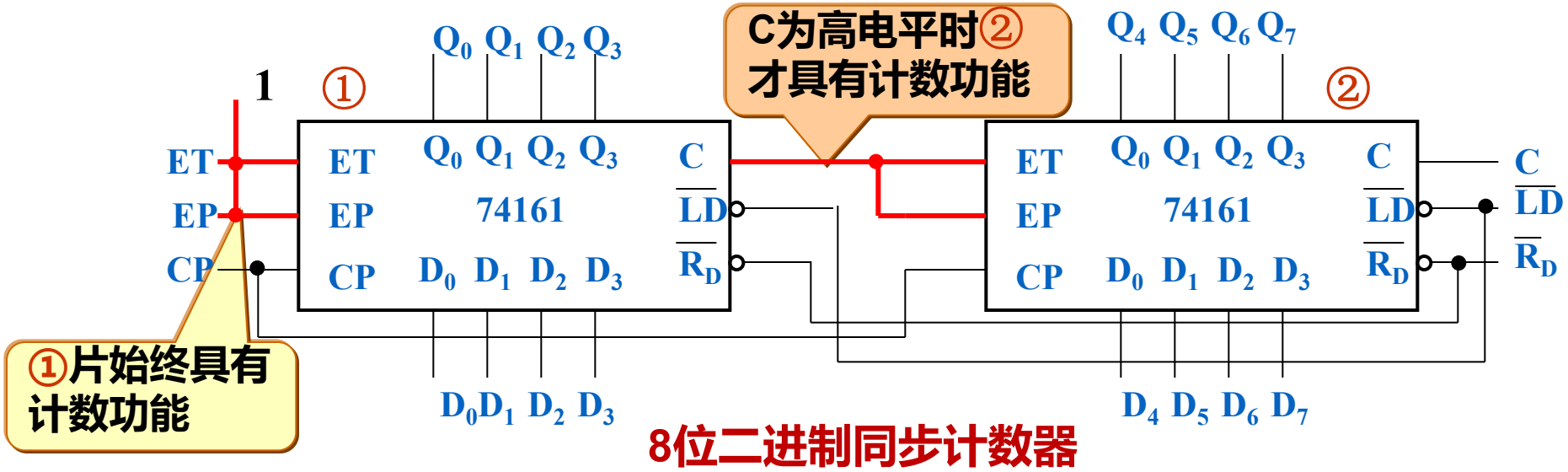


串行进位法



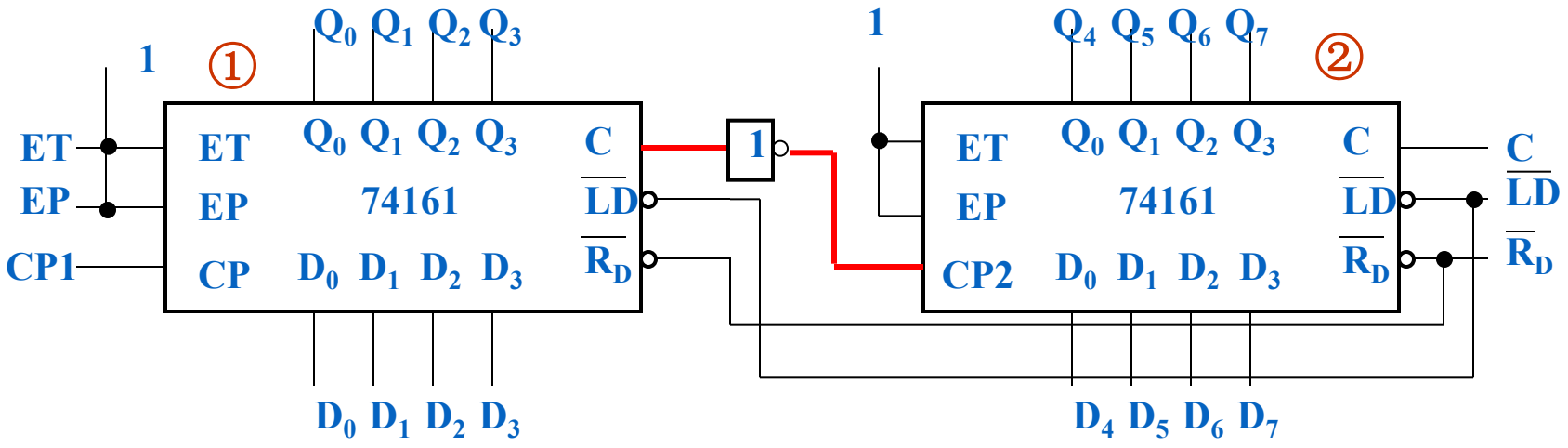
集成计数器的同步扩展

将若干片集成计数器级联起来，形成有较大模值的计数系统。



- 同步扩展指所有计数器使用同一个时钟信号来同步，即把两片4位二进制计数器的CP并联后接时钟信号CP。
- 低位片的ET、EP接高电平，使低位片始终具有计数功能；高位片的ET、EP接低位片的进位输出端C，只有当C为高电平时，高位片才具有计数功能。

集成计数器的异步扩展



- 异步指多个计数器不是统一由一个时钟信号来同步的，各自有单独的时钟。
- 低位片4位二进制计数器的CP接系统时钟信号CP1，低位片的进位输出端C经反相后接高位片的CP端。

三、任意进制计数器的构成方法

2. N < M

②M不可分解

采用整体置零和整体置数法：

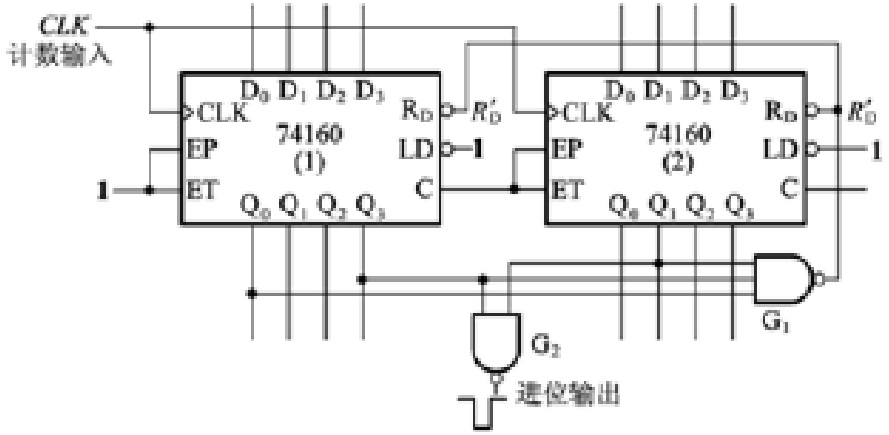
先用两片接成 M'> M 的计数器

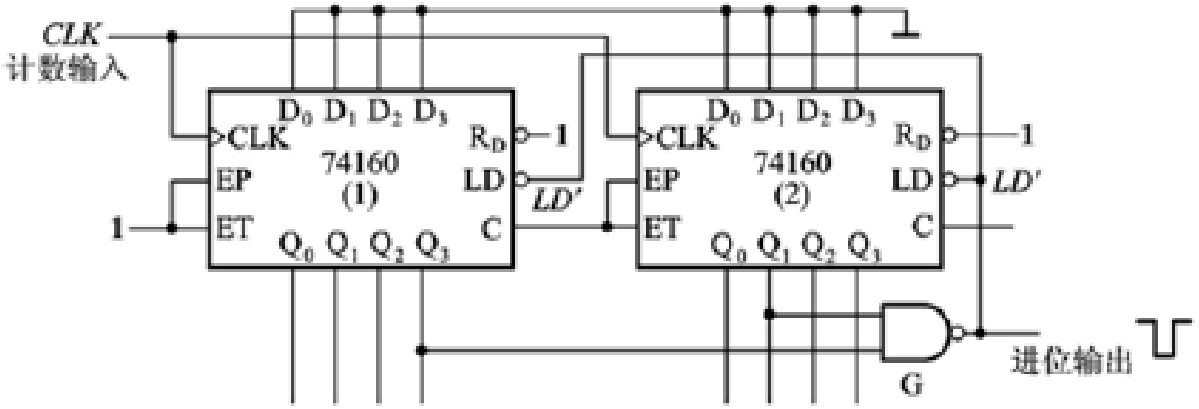
然后再采用置零或置数的方法

例：用74160接成29进制

整体置零

（异步）



整体置数

（同步）